

Efficient Estimation for Network-Linked High-Dimensional Multinomial Probit via Penalized Pseudo Likelihood

Rashmi Ranjan Bhuyan* Adel Javanmard* Gourab Mukherjee*

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Abstract

The multinomial probit model (MNP) is widely used for analyzing unordered categorical data. In a host of contemporary applications, we encounter cross-sectional datasets with categorical responses and high-dimensional covariates. In absence of repeated observations from the respondents, MNP models in these applications are often equipped with additional network structures that leverage contiguity in the responses of similar units. Pooling information across similar units through these network structures can provide significantly better inference in these data scarcity problems. However, estimating the effects of sparse high-dimensional covariates in the presence of network linkages among the responses is challenging. We develop a novel Monte Carlo EM algorithm that exploits the pseudo-likelihood of the problem and estimates the covariate effects as well as the network strength. We provide decision theoretic guarantees on the operational characteristic of the pseudo-likelihood. We demonstrate the application of the proposed method in spatial autocorrelation network structured MNP models. We compare the performance of the proposed algorithm on a wide range of simulation experiments and obtain encouraging performance. Finally, an application to Facebook page classification data illustrates how the pseudo-likelihood approach delivers interpretable estimates and better out-of-sample prediction when both covariate sparsity and network spillovers are present.

Keywords: Pseudo-Likelihood, Network-Linked, Multinomial Probit, High Dimensional

*Data Science and Operations Department, Marshall School of Business, University of Southern California

1 Introduction

Modeling choices, where an individual has two or more discrete (i.e. distinct and separable; mutually exclusive) alternatives to choose from, is a central problem in statistics and especially econometrics. A widely used framework for these problems is the discrete choice model. Discrete choice models operate within the broader context of rational choice theory. The underlying assumption is that an individual assigns a utility to each available option and selects the one that provides the highest utility.

The multinomial probit model (MNP) is one such popular discrete choice model used for analyzing unordered categorical data. The probit model assumes that for any individual the utility is defined as a linear combination of covariates and a Gaussian noise and based on these utilities a choice is made. There have been works in using Bayesian techniques (McCulloch and Rossi, 1994; McCulloch et al., 2000) to estimate the parameters in such model. Most of the works assume that such observations of choices and the corresponding utilities are independent. However, it is well known that there exists a “peer effect” in such choices. Basically, the people socially connected to an individual do influence their utility for the choices and hence the choice they eventually make. In such settings the observations are not independent and existing models fail to incorporate these structures.

Spatial models provide a natural way to model this “peer effect” between different individuals. Such structures have been generally modeled with the spatial autoregressive models (SAR) and the spatial error models (SEM). These structures can be social networks, proximity based networks or any other distance based on covariates. This presents a challenging problem since each individual is unique based on their network connection, and there is a lack of repeated observations for each individual.

The challenges of the spatial dependence structures are compounded by modern datasets because of the existence of high dimensional covariates. Variable selection becomes an important part of the problem here since existence of the nuisance covariates increases the estimation errors. In this work, we define a spatially structured multinomial probit model. We then provide a framework for variable selection in such a model. Later, we also present extensions of our method to alternative network structures.

1.1 Our contributions and connections to existing works

We address the problem of variable selection in the SAR multinomial probit framework. The likelihood of the probit model, combined with the dependence structure imposed by the SAR framework, makes an analytical solution impossible. Additionally, the likelihood is computationally intensive and becomes infeasible to compute in higher dimensions, which means that gradient-based optimization techniques cannot be used. Since the model has

inherent latent variables, the Expectation-Maximization (EM) algorithm is a standard technique that can be applied. However, even this approach fails because computing the expectation in the E-step is extremely difficult. As a result, standard solutions for the SAR multinomial probit model either use a Monte Carlo EM algorithm or Bayesian methods. Bayesian methods are generally computationally intensive and also face the additional challenge of prior specification. The practical implementation of MCEM is also difficult, as it requires sampling from a high-dimensional truncated normal distribution.

We propose Pseudo-Likelihood based EM (PLEM) as a solution to the spatial probit model. The pseudo-likelihood based approach with an added l_1 penalty on the high dimensional parameters ensures that our method produces sparse solution and are computationally efficient. We show that the solution obtained from PLEM does converge to the true minimizer under certain conditions. With our extensive numerical experiments we also show that our proposed PLEM recovers the signals in our sparse parameter without selecting many nuisance variables.

Our research builds on works in discrete choice modeling, high dimensional variable selection and spatial models . Next, we list the relevant literature in these research and also briefly mention our contributions.

1) Discrete choice modeling. Discrete choice models are popular frameworks in econometrics where a consumer makes a decision based on their utility of different choices. [McFadden et al. \(1973\)](#) introduced the multinomial logit (MNL) to analyze the decision of choosing the mode of transport for shopping trips. Within this choice modeling framework [Hausman and Wise \(1978\)](#) formulated the multinomial Probit model. Unlike the multinomial logit model, the probit model allows correlation between the probability of various choices and hence is preferable in a variety of scenarios. Even with the benefits, the Probit model is less popular because of the computational complexities stemming from the intractable likelihood. [McCulloch and Rossi \(1994\)](#) provide an analysis of the likelihood function and a Bayesian estimation of the parameters by devising a Gibbs's sample. The other popular approach to the estimation of MNP is the Monte Carlo EM algorithm ([Natarajan et al., 2000](#)). We extend on the MNP model by presenting SAR MNP which allows for correlation across observations and develop NPEM, a Monte Carlo EM based approach for estimation.

2) Spatial models and applications. Spatial models provide a natural way to model dependence structure between different observations based on how close they are in some space. Especially for discrete choice models, [LeSage and Pace \(2009\)](#) use the spatial autoregressive (SAR) models with Probit and Tobit regression to introduce correlation across choices. SAR models have been successful in using spatial infor-

mation in many real-world datasets (Manski, 1993; Anselin and Arribas-Bel, 2013). Though called spatial model, any network setting can be incorporated into an SAR structure, Hsieh and Lee (2016) used SAR with a friendship networks at high school to predict academic performance. Specifically, the SAR probit has been used to model land development decisions for undeveloped parcels in Austin, Texas (Wang et al., 2014). Based on this existing literature, the SAR model can well capture the dependence among observations in real-world data.

3) High dimensional data. With the advent of online data collection, there are numerous occasions where there is an abundance of data and we encounter high dimensional features. Ideally, only a few of these covariates actually effect the outcome we are trying to model. A sparse coefficient corresponding to the covariates performs as variable selection method. It is common to use the notion of shrinkage to estimate such sparse parameters as it often leads to substantial improvements in performances. The lasso (Tibshirani, 1996) is the most popular method for such shrinkage that uses an l_1 penalty to achieve a sparse solution. This idea of using an l_1 penalty to introduce sparsity has been applied to generalized linear models (Tibshirani, 1996), time series (Valdés-Sosa et al., 2005; Hsu et al., 2008) as well as wavelet analysis (Donoho and Johnstone, 1994; Antoniadis and Fan, 2001). Compared to most such models which assume independence across observations, Mukherjee et al. (2024) Adel: this has been published, update the bibliography. Also check other arXiv references to make sure they are updated. Changed and updated others. use this l_1 penalty in logistic regression under network dependence. We develop a method to estimate high dimensional parameters in a multinomial Probit model under network dependence by inducing an l_1 penalty.

1.2 Notation

Section added Throughout the paper, we use the following notation. For a vector $\mathbf{v} = (v_1, \dots, v_d) \in \mathbb{R}^d$ and any $0 < q < \infty$ The ℓ_q -norm of $\mathbf{v}$ is $\|\mathbf{v}\|_q = \left(\sum_{k=1}^d |v_k|^q\right)^{1/q}$, and the maximum norm is $\|\mathbf{v}\|_\infty = \max_{1 \leq k \leq d} |v_k|$. We also define $\|\mathbf{v}\|_0 = \sum_{k=1}^d \mathbf{1}\{v_k \neq 0\}$, the ‘zero-norm’, which counts the number of non-zero entries of $\mathbf{v}$.

For a matrix $\mathbf{A} = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n} \in \mathbb{R}^{m \times n}$, we define $\|\mathbf{A}\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|$, the row-wise maximum ℓ_1 norm.

1.3 Organization of this paper

The rest of this paper is organized as follows. We begin with a background on the multinomial probit model under network dependence and introduce some notations in Section 2. In Section 3 we introduce the pseudo-likelihood framework for variable selection as well as PLEM, while Section 4 presents a theoretical analysis of the proposed framework. The empirical performance of PLEM is investigated in Sections 5 and 6. Proofs, and other technical details are relegated to the appendix.

2 SAR Multinomial probit model

We are given observations $(y_i, \mathbf{x}_{1i}, \mathbf{x}_{2i}, \dots, \mathbf{x}_{Ji}) \in \{0, 1, 2, \dots, J\} \times \mathbb{R}^p \times \mathbb{R}^p \times \dots \times \mathbb{R}^p$ for $i = 1, \dots, n$ and consider the setting where the response vector $\mathbf{y} = (y_1, \dots, y_n)$ represents n nodes on a network. At each node i , suppose there are *latent variables* $y_{1i}, y_{2i}, \dots, y_{Ji}$ such that,

$$y_{ji} = \rho \sum_{p \neq i}^n W_{ip} y_{jp} + \mathbf{x}_{ji}^T \boldsymbol{\beta}_j + \epsilon_{ji}, \quad (\epsilon_{1i}, \epsilon_{2i}, \dots, \epsilon_{Ji})^\top \stackrel{i.i.d}{\sim} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \quad (1)$$

where $\mathbf{W} = (W_{ij})$ is a known $n \times n$ symmetric interaction matrix with 0 in the diagonal and $\boldsymbol{\Sigma}$ denotes the covariance across choices. For identifiability the first diagonal element of $\boldsymbol{\Sigma}$ is assumed to be 1. Based on these latent variables, the observation y_i is then described as class 0 if all of the latent variables are negative or j^* if $y_{j^*i} = \max\{y_{1i}, y_{2i}, \dots, y_{Ji}\} > 0$. Equation (1) combined with the definition of y_i based on the latent variables represent the classical SAR multinomial probit model (see Sec.10.4 of [LeSage and Pace \(2009\)](#)).

Denote $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^T, \boldsymbol{\beta}_2^T, \dots, \boldsymbol{\beta}_J^T)^T$ to be the $pJ \times 1$ column vector of the regression parameters, $\tilde{\mathbf{W}} = \mathbf{W} \otimes \mathbf{I}_J$ the $nJ \times nJ$ rearranged network matrix, $\tilde{\mathbf{X}}$ to be the corresponding $nJ \times pJ$ matrix of predictors defined as $\tilde{\mathbf{X}} = (\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n)^\top$, where $\mathbf{X}_i$ is a block diagonal matrix constructed using $\mathbf{x}_{1i}, \mathbf{x}_{2i}, \dots, \mathbf{x}_{Ji}$, covariates for observation i corresponding to each alternative. Next, define $\tilde{\mathbf{y}}$, as $\tilde{\mathbf{y}} = (\tilde{\mathbf{y}}_1, \tilde{\mathbf{y}}_2, \dots, \tilde{\mathbf{y}}_n)$ where $\tilde{\mathbf{y}}_i = (y_{1i}, y_{2i}, \dots, y_{Ji})$ and similarly define $\tilde{\boldsymbol{\epsilon}}$. With these notations, the latent variables from (1) can be rewritten as

$$\mathbf{H} \tilde{\mathbf{y}} = \tilde{\mathbf{X}} \boldsymbol{\beta} + \tilde{\boldsymbol{\epsilon}}, \quad \tilde{\boldsymbol{\epsilon}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n \otimes \boldsymbol{\Sigma}), \quad (2)$$

where $\mathbf{H} := \mathbf{I}_{nJ} - \rho \tilde{\mathbf{W}}$. Using (2), see that $\tilde{\mathbf{y}} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Omega})$ where $\boldsymbol{\mu} = \mathbf{H}^{-1} \tilde{\mathbf{X}} \boldsymbol{\beta}$ and $\boldsymbol{\Omega} = (\mathbf{H})^{-1} (\mathbf{I}_n \otimes \boldsymbol{\Sigma}) (\mathbf{H}^T)^{-1}$. To describe the likelihood of SAR probit model, we need first describe some linear constraints corresponding to the value of y_i . We construct a matrix of linear constraints $\mathbf{D}(\mathbf{y})$ based on the response $\mathbf{y}$ such that $\mathbf{D}(\mathbf{y}) \tilde{\mathbf{y}} \leq \mathbf{0}_{nJ}$. The construction

is as follows

$$\mathbf{D}(\mathbf{y}) = \begin{pmatrix} \mathbf{D}^{(y_1)} & \mathbf{0}_J & \dots & \mathbf{0}_J \\ \mathbf{0}_J & \mathbf{D}^{(y_2)} & & \vdots \\ \vdots & & \ddots & \mathbf{0}_J \\ \mathbf{0}_J & \dots & \mathbf{0}_J & \mathbf{D}^{(y_n)} \end{pmatrix}$$

where $\mathbf{D}^{(0)} = \mathbf{I}_J$ and for $j \geq 1$, $\mathbf{D}^{(j)}$ is the $J \times J$ identity matrix with the j^{th} column replaced with -1 . With the construction of $\mathbf{D}(\mathbf{y})$, we can define the likelihood $\ell(\boldsymbol{\beta}, \boldsymbol{\Sigma}, \rho) = \int_{\mathcal{A}} \phi_{nJ}(\tilde{\mathbf{y}}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\mathcal{A} = \{\tilde{\mathbf{y}} : \mathbf{D}(\mathbf{y})\tilde{\mathbf{y}} \leq \mathbf{0}_{nJ}\}$, $\boldsymbol{\mu} = \mathbf{H}^{-1}\tilde{\mathbf{X}}\boldsymbol{\beta}$ and $\boldsymbol{\Omega} = (\mathbf{H})^{-1}(\mathbf{I}_n \otimes \boldsymbol{\Sigma})(\mathbf{H}^T)^{-1}$. We will assume that $\boldsymbol{\beta}_j$ are sparse and our goal is to estimate $\boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\Sigma}, \rho)$. A natural approach is to minimize the following objective function with respect to $\boldsymbol{\theta}$,

$$\min_{\boldsymbol{\theta}} -\ell(\boldsymbol{\beta}, \boldsymbol{\Sigma}, \rho) + \lambda \sum_{j=1}^J \|\boldsymbol{\beta}_j\|_1. \quad (3)$$

2.1 Monte-Carlo Expectation Maximization

While (3) provides a natural way to estimate the parameters in our model, an analytical solution does not exist and we need to resort to numerical solutions. But the likelihood for our model is integral of a multivariate normal distribution over an affine space $\mathcal{A}$, where we recall $\mathcal{A} = \{\tilde{\mathbf{y}} : \mathbf{D}(\mathbf{y})\tilde{\mathbf{y}} \leq \mathbf{0}_{nJ}\}$. Hence, it does not have a closed form solution. Even computing the likelihood is extremely hard for large n and J . As the number of choices J increases, the size of the affine space $\mathcal{A}$ decreases and with increasing n and J the dimension of the multivariate normal also increases. This makes the computation of the likelihood, even for a fixed set of parameters, difficult. Any optimization method would in the least require the computation of this likelihood several times in order to minimize (3) which is hard.

Adel: Expand this to better explain the EM algorithm, emphasize what is latent, what is observed in our model and interpretation This problem is generally tackled with the Expectation Maximization (EM) algorithm by utilizing the inherent latent variable structure of the problem. The E-step in the EM algorithm computes the expected value of $f(\boldsymbol{\theta}|\tilde{\mathbf{X}}, \mathbf{y})$ Adel: What is f ?, given the observed data and the current parameter estimate $\boldsymbol{\theta}^{(t-1)}$. Define $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t-1)}) := \mathbb{E}[f(\boldsymbol{\theta}|\tilde{\mathbf{X}}, \mathbf{y})|\boldsymbol{\theta}^{(t-1)}]$ Adel: Write it explicitly that the expectation is with respect to But to get the expectation in the Q function, we need to compute $\int l(\boldsymbol{\theta}|\tilde{\mathbf{X}}, \tilde{\mathbf{y}})\mathbb{P}(\tilde{\mathbf{y}}|\tilde{\mathbf{X}}, \boldsymbol{\theta}^{(t-1)}, \mathbf{y})d\mathbf{y}$, where $\mathbb{P}(\tilde{\mathbf{y}}|\tilde{\mathbf{X}}, \boldsymbol{\theta}^{(t-1)}, \mathbf{y})$ follows a truncated normal distribution with mean $\mathbf{H}^{-1}\tilde{\mathbf{X}}\boldsymbol{\beta}^{(t-1)}$ and covariance $(\mathbf{H})^{-1}(\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{(t-1)})(\mathbf{H}^T)^{-1}$. Adel: Why? I agree $\mathbb{P}(\tilde{\mathbf{y}}|\tilde{\mathbf{X}}, \boldsymbol{\theta}^{(t-1)})$ is normal with this mean and covariance, but why conditioning on $\mathbf{y}$ does not alter the mean and covariance and only makes it truncated? Also is it truncated or projected?

It is truncated normal, not projected normal. The truncated multivariate normal distribution with mean μ and variance Σ with truncation implies that, if $y \sim TMVN(\mu, \Sigma)$, with truncation such that $y \in \mathcal{A}$, then density of y is given by $\phi(\mu, \Sigma)/\mathbb{P}(x \in \mathcal{A})$ if $y \in \mathcal{A}$, 0 otherwise, where $x \sim \mathcal{N}(\mu, \Sigma)$. In this case since conditioning on $\mathbf{y}$ only tells us the sign, we just know that it belongs in a specific region thus mean and variance parameters remain the same. I can either cite this from Chapter 10.1.1, Intro to spatial econometrics book or explain it in detail. Added in details below.

Our observed data is $(\tilde{\mathbf{X}}, \mathbf{y})$, the covariates and the realized choices, whereas the latent vector is $\tilde{\mathbf{y}} = (\tilde{y}_1, \dots, \tilde{y}_n) \in \mathbb{R}^{nJ}$ and $\tilde{\mathbf{y}}_i = (y_{1i}, y_{2i}, \dots, y_{Ji})$. Here, each component $\tilde{y}_{ji}$ represents the (unobserved) utility of alternative j for unit i so we do not see these utilities directly, instead, we observe only the selected alternative y_i . This problem is generally tackled with the Expectation Maximization (EM) algorithm by utilizing the inherent latent variable structure of the problem. We first define the penalized loss function $f(\boldsymbol{\theta}|\tilde{\mathbf{X}}, \mathbf{y}) := -\ell(\boldsymbol{\beta}, \boldsymbol{\Sigma}, \rho|\tilde{\mathbf{X}}, \mathbf{y}) + \lambda\|\boldsymbol{\beta}\|_1$, which we intend to minimize with respect to $\boldsymbol{\theta}$. In the E-step, we take the conditional expectation of this objective given the observed data and the parameter estimate of the current iterate, thereby defining $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t-1)}) := \mathbb{E}[f(\boldsymbol{\theta}|\tilde{\mathbf{X}}, \mathbf{y})|\boldsymbol{\theta}^{(t-1)}]$, the expectation of $f(\boldsymbol{\theta}|\tilde{\mathbf{X}}, \mathbf{y})$, the penalized loss function conditioned on the current estimate of parameter $\boldsymbol{\theta}^{(t-1)}$.

To compute $Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t-1)})$, we need the conditional expectation $\mathbb{E}[f(\boldsymbol{\theta}|\tilde{\mathbf{X}}, \mathbf{y})|\boldsymbol{\theta}^{(t-1)}]$ which necessitates computing the conditional expectation of $\ell(\boldsymbol{\beta}, \boldsymbol{\Sigma}, \rho|\tilde{\mathbf{X}}, \mathbf{y})$ given parameter estimate $\boldsymbol{\theta}^{(t-1)}$. This is simplified as $\int l(\boldsymbol{\theta}|\tilde{\mathbf{X}}, \tilde{\mathbf{y}})\mathbb{P}(\tilde{\mathbf{y}}|\tilde{\mathbf{X}}, \boldsymbol{\theta}^{(t-1)}, \mathbf{y})d\mathbf{y}$ where $\mathbb{P}(\tilde{\mathbf{y}}|\tilde{\mathbf{X}}, \boldsymbol{\theta}^{(t-1)}, \mathbf{y})$ is the distribution of the latent values $\tilde{\mathbf{y}}$ conditional on data $(\tilde{\mathbf{X}}, \mathbf{y})$ and current estimate of parameters $\boldsymbol{\theta}^{(t-1)}$.

Definition 1. Let $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and let $\mathcal{A} \subset \mathbb{R}^d$ be a set such that $\mathbb{P}(\mathbf{x} \in \mathcal{A}) > 0$. The truncated multivariate normal (TMVN) distribution on $\mathcal{A}$ is the distribution of $\mathbf{y} := \mathbf{x} \mid (\mathbf{x} \in \mathcal{A})$, denoted as $\mathbf{y} \sim \text{TMVN}(\boldsymbol{\mu}, \boldsymbol{\Sigma}; \mathcal{A})$ and its density is given by

$$f(\mathbf{y}) = \begin{cases} \frac{\phi_d(\mathbf{y}; \boldsymbol{\mu}, \boldsymbol{\Sigma})}{\mathbb{P}(\mathbf{x} \in \mathcal{A})} = \frac{\phi_d(\mathbf{y}; \boldsymbol{\mu}, \boldsymbol{\Sigma})}{\int_{\mathcal{A}} \phi_d(\mathbf{u}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) d\mathbf{u}}, & \mathbf{y} \in \mathcal{A}, \\ 0, & \mathbf{y} \notin \mathcal{A}, \end{cases}$$

where $\phi_d(\mathbf{y}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ is the density of a d -dimensional Gaussian random variable with parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

As noted in Chapter 10.1.1 in [LeSage and Pace \(2009\)](#), the conditional distribution of latent variables $\tilde{\mathbf{y}}$ is given by *truncated multivariate normal* (TMVN) distribution, specifically,

$$\mathbb{P}(\tilde{\mathbf{y}} | \tilde{\mathbf{X}}, \boldsymbol{\theta}^{(t-1)}, \mathbf{y}) \sim \text{TMVN}(\mathbf{H}^{-1} \tilde{\mathbf{X}} \boldsymbol{\beta}^{(t-1)}, (\mathbf{H})^{-1} (\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{(t-1)}) (\mathbf{H}^T)^{-1}).$$

Since the analytical solution to the expected value of a truncated normal random variable does not exist, the expectation $\mathbb{E}[f(\boldsymbol{\theta} | \tilde{\mathbf{X}}, \mathbf{y}) | \boldsymbol{\theta}^{(t-1)}]$ is intractable and thus we need to use a Monte-Carlo approximation to the expectation in this setting. This means that the Q function is approximated as $Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(t-1)}) \approx \frac{1}{K} \sum_{k=1}^K l(\boldsymbol{\theta} | \tilde{\mathbf{X}}, \tilde{\mathbf{y}}_k^{(t)}) + \lambda \|\boldsymbol{\beta}\|_1$ where $\tilde{\mathbf{y}}_k^{(t)} \sim \mathbb{P}(\tilde{\mathbf{y}} | \tilde{\mathbf{X}}, \boldsymbol{\theta}^{(t-1)}, \mathbf{y})$.

Once we have the Q function from the E-step, the M-step then consists of minimizing the loss computed in the E-step, as follows:

$$\boldsymbol{\theta}^{(t)} := \min_{\boldsymbol{\theta}} Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(t-1)}).$$

We do this in a three-stage conditional minimization.

- Stage 1: $\boldsymbol{\beta}^{(t)} := \min_{\boldsymbol{\beta}} Q(\boldsymbol{\beta} | \boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{(t-1)}, \rho = \rho^{(t-1)})$
- Stage 2: $\boldsymbol{\Sigma}^{(t)} := \min_{\boldsymbol{\Sigma}} Q(\boldsymbol{\Sigma} | \boldsymbol{\beta} = \boldsymbol{\beta}^{(t)}, \rho = \rho^{(t-1)})$
- Stage 3: $\rho^{(t)} := \min_{\rho} Q(\rho | \boldsymbol{\beta} = \boldsymbol{\beta}^{(t)}, \boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{(t)})$

We provide the details of the three stages and their minimization criteria in the appendix. Algorithm 1 describes the MCEM solution to the high-dimensional SAR probit model.

2.2 MCEM Computation Challenges

While MCEM provides a solution to our optimization problem, it is not computationally viable. To use Algorithm 1, we generate samples from the truncated multivariate normal of size nJ . ~~Since the latent variables are correlated across alternatives and observations~~

Algorithm 1 Monte-Carlo Expectation Maximization for High-Dimensional SAR Probit

Data : $\mathbf{W}$ - interaction matrix, $\mathbf{X}$ - covariate matrix, $\mathbf{y}$ - response vector

Initialize Set $\beta^{(0)} = \beta_0$, $\Sigma^{(0)} = \Sigma_0$, $\rho^{(0)} = \rho_0$.

for $t = 1, 2, \dots, T$ **do**

MC E-Step: Generate samples of the unknown latent variable $\tilde{\mathbf{Y}}$, using the current value of parameter $\beta^{(t-1)}$, $\Sigma^{(t-1)}$, $\rho^{(t-1)}$:

- Set $\mathbf{H}^{(t)} = \mathbf{I}_{nJ} - \rho^{(t-1)}\tilde{\mathbf{W}}$, $\mu^{(t)} = (\mathbf{H}^{(t)})^{-1}\tilde{\mathbf{X}}\beta^{(t-1)}$ and $\Omega^{(t)} = (\mathbf{H}^{(t)})^{-1}(\mathbf{I}_n \otimes \Sigma^{(t-1)})((\mathbf{H}^{(t)})^T)^{-1}$
- Generate K samples $\tilde{\mathbf{y}}_k^{(t)} \stackrel{i.i.d}{\sim} \text{TMVN}(\mu^{(t)}, \Omega^{(t)}); \mathbf{D}(\mathbf{y})\tilde{\mathbf{y}}_k^{(t)} \leq \mathbf{0}$ **Adel: What is K ? How to choose it?** K is the number of samples used to estimate the expectation for the E step. I don't have a recommendation of what K should be. I have used $K = 10$ for all simulations and real data analyses.
- Compute the mean to get $\tilde{\mathbf{y}}^{(t)} = \sum_{k=1}^K \tilde{\mathbf{y}}_k^{(t)} / K$
- Motivated by (2), define $\tilde{\mathbf{y}}^{*(t)} := \mathbf{H}^{(t)}\tilde{\mathbf{y}}^{(t)}$, $\tilde{\mathbf{y}}_k^{*(t)} := \mathbf{H}^{(t)}\tilde{\mathbf{y}}_k^{(t)}$

M-Step (a): Conditional maximization of β

- Estimate $\beta^{(t)}$ by performing Lasso with $(\mathbf{I}_n \otimes \Sigma^{-1/2})\tilde{\mathbf{y}}^{*(t)}$ and $(\mathbf{I}_n \otimes \Sigma^{-1/2})\tilde{\mathbf{X}}$

M-Step (b): Conditional maximization of Σ

- Define errors for w.r.t the samples from the E-Step and the estimates of β from the M-Step(a) as $\epsilon_{jik}^{(t)} := \tilde{y}_{kji}^{*(t)} - \mathbf{x}_{ji}\beta_j^{(t)}$
- Let $\mathbf{E}^{(t)} = \left(\sum_{k=1}^K \sum_{i=1}^n \tilde{\epsilon}_{ik}^{(t)} \tilde{\epsilon}_{ik}^{(t)T} \right) / nK$ where $\tilde{\epsilon}_{ik}^{(t)} = (\epsilon_{1ik}^{(t)}, \epsilon_{2ik}^{(t)} \dots, \epsilon_{Jik}^{(t)})^T$
- Partition the sample covariance matrix as $\mathbf{E}^{(t)} = \left[\begin{array}{c|c} (\mathbf{E}_1^{(t)})_{1 \times 1} & (\mathbf{E}_2^{(t)})_{1 \times (J-1)} \\ \hline (\mathbf{E}_3^{(t)})_{(J-1) \times 1} & (\mathbf{E}_4^{(t)})_{(J-1) \times (J-1)} \end{array} \right]$
- $\boldsymbol{\eta}^{(t)} = \mathbf{E}_3^{(t)} / \mathbf{E}_1^{(t)}$, $\boldsymbol{\Phi}^{(t)} = \mathbf{E}_4^{(t)} - \mathbf{E}_1^{(t)}\boldsymbol{\eta}^{(t)}\boldsymbol{\eta}^{(t)T}$ and $\Sigma^{(t)} = \left[\begin{array}{c|c} 1 & \boldsymbol{\eta}^{(t)T} \\ \hline \boldsymbol{\eta}^{(t)} & \boldsymbol{\Phi}^{(t)} + \boldsymbol{\eta}^{(t)}\boldsymbol{\eta}^{(t)T} \end{array} \right]$

M-Step (c): Conditional maximization of ρ

- Estimate $\rho^{(t)}$ by maximizing conditional on $\beta^{(t)}$ and $\Sigma^{(t)}$ **Adel: This is scalar optimization, but need to give more details on how you do the maximization there is analytical solution to it. It is included in the appendix. Also adding it here. Please let me know if I should keep it here or just refer to appendix.**

$$\rho^{(t)} = \frac{\sum_{k=1}^K \left(\tilde{\mathbf{y}}_k^{(t)} \right)^T (\mathbf{I}_n \otimes \Sigma^{-1}) \tilde{\mathbf{W}} \tilde{\mathbf{y}}_k^{(t)} - K \left((\tilde{\mathbf{X}}\beta^{(t)})^T (\mathbf{I}_n \otimes \Sigma^{-1}) \tilde{\mathbf{W}} \tilde{\mathbf{y}}^{(t)} \right)}{\sum_{k=1}^K \left(\tilde{\mathbf{y}}_k^{(t)} \right)^T \tilde{\mathbf{W}}^T (\mathbf{I}_n \otimes \Sigma^{-1}) \tilde{\mathbf{W}} \tilde{\mathbf{y}}_k^{(t)}}.$$

Output $\beta_1^{(T)}, \beta_2^{(T)}, \dots, \beta_J^{(T)}$, $\Sigma^{(T)}$ and $\rho^{(T)}$

Adel: What are alternatives, what are observations? alternatives are the J choices, observations are the n observations. Changed the sentence to better describe. Since the latent variables are correlated across J choices and the n observations, this is not a separable problem, thus we need to sample from the nJ dimensional truncated normal distribution. Rejection sampling is the major way to sample from the truncated normal distribution. To compute complexity of generating one sample, we need to compute the probability of acceptance for the rejection sampling. Finding the analytical solution to the probability is not feasible. Rather, we will provide an approximate relation of such probability. The samples we need to generate, $\tilde{\mathbf{y}} \stackrel{i.i.d}{\sim} \text{TMVN}(\boldsymbol{\mu}, \boldsymbol{\Omega})$ such that $\mathbf{D}(\mathbf{y})\tilde{\mathbf{y}} \leq \mathbf{0}$. This means, using rejection sampling, we generate samples from $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Omega})$ and then accept a sample $\tilde{\mathbf{y}}$ if $\mathbf{D}(\mathbf{y})\tilde{\mathbf{y}} \leq \mathbf{0}$.

To simplify, consider the corresponding counterpart, $\bar{\mathbf{y}} \sim \mathcal{N}(\mathbf{D}(\mathbf{y})\boldsymbol{\mu}, \mathbf{D}(\mathbf{y})\boldsymbol{\Omega}\mathbf{D}(\mathbf{y})^T)$, and accept the sample $\bar{\mathbf{y}}$ if $\bar{\mathbf{y}} \leq \mathbf{0}$. This is the same problem as the previous one with just the transformed variable $\bar{\mathbf{y}} = \mathbf{D}(\mathbf{y})\tilde{\mathbf{y}}$. Hence the acceptance probability is same as the probability of the orthant of $\bar{\mathbf{y}}$ such that $\bar{\mathbf{y}} \leq \mathbf{0}$. In the very simple case that $\bar{\mathbf{y}}$ is standard Gaussian, then this orthant probability would be $(1/2)^{nJ}$ since $\bar{\mathbf{y}}$ is of dimension nJ . Hence, the expected number of samples required to generate a single sample is 2^{nJ} and the complexity is $\mathcal{O}(2^{nJ})$.

This is under the simplifying assumption that the orthant probabilities of $\bar{\mathbf{y}}$ behave the same way as standard Gaussian distribution. We also provide a numerical simulation to show the exponential nature of the acceptance probabilities. We set the number of choices J as 2 and vary n from 50 to 200 and record the probability of acceptance across the different n 's. We then compute the orthant probabilities after doing the required transformation. In Figure 1(a), we plot the log probabilities of the orthants against various values of n . We simultaneously plot a line with intercept 0 and slope $-1/6$ and see a linear trend of the $\log(\text{probability})$ which strengthens our belief that the acceptance probabilities are indeed exponentially decreasing and the complexity is exponential in n . In Figure 1(b), we fix n as 100 and vary the number of choices J from 2 to 10 and plot the log probabilities of the orthants. We see a similar linear trend implying that the acceptance probabilities are exponentially decreasing as well and the complexity is exponential in J as well. Adel: How about a similar plot, where you fix n and vary J ? I have added a plot with varying J as well.

3 Proposed PLEM Algorithm

In the previous section, we introduced the MCEM algorithm to estimate the parameter in the high-dimensional SAR probit model. As we noted, while it can estimate the model, it is

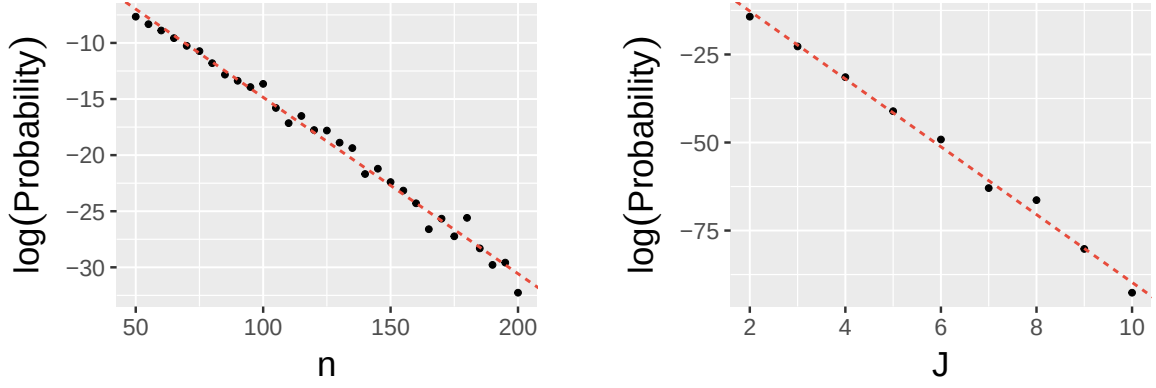


Figure 1: $\log \mathbb{P}(\bar{\mathbf{y}} \leq \mathbf{0})$ versus dimension: (a) $J = 2$ with n varying; (b) $n = 100$ with J varying.

computationally extremely expensive. The major issue in the formulation is the correlation across the n observations. This crates the intricate correlation structure across all the nJ latent variables which in turn creates the bottleneck in the Monte-Carlo E-step. Rather than using the full distribution of the data, in this section we introduce a Pseudo-Likelihood based approach that only takes into account the marginal distribution of y_i instead of the joint distribution of $\mathbf{y}$. Let $\mathbf{H} = \mathbf{I}_{nJ} - \rho \tilde{\mathbf{W}}$ and recall that the latent-variable representation in (2) implies $\tilde{\mathbf{y}} = (\tilde{\mathbf{y}}_1^\top, \tilde{\mathbf{y}}_2^\top, \dots, \tilde{\mathbf{y}}_n^\top)^\top \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Omega})$ where $\boldsymbol{\mu} = \mathbf{H}^{-1} \tilde{\mathbf{X}} \boldsymbol{\beta}$ and $\boldsymbol{\Omega} = \mathbf{H}^{-1} (\mathbf{I}_n \otimes \boldsymbol{\Sigma}) \mathbf{H}^{-\top}$.

For any matrix $\mathbf{M} \in \mathbb{R}^{nJ \times nJ}$, denote by $\mathbf{M}_{ii}$ its i -th $J \times J$ diagonal block and by $\mathbf{M}_i$ (or $(\mathbf{M})_i$) the i -th J -vector block row. Then for all $i = 1, \dots, n$, $\tilde{\mathbf{y}}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Omega}_{ii})$. Finally, for each observation, we define the plausible region based on the observation y_i as $A_i = \{\tilde{\mathbf{y}}_i \in \mathbb{R}^J : \mathbf{D}^{(y_i)} \tilde{\mathbf{y}}_i \leq \mathbf{0}_J\}$, where $\mathbf{D}^{(y_i)}$ is the $J \times J$ matrix as defined earlier. The probability of observing y_i is therefore

$$p_i(\boldsymbol{\theta}) = \int_{A_i} \phi_J(\tilde{\mathbf{y}}_i \mid \boldsymbol{\mu}_i, \boldsymbol{\Omega}_{ii}) d\tilde{\mathbf{y}}_i, \quad \boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\Sigma}, \rho). \quad (4)$$

The pseudo-likelihood is then defined by combining the marginal (node-wise) pseudo-log-likelihood $\tilde{\ell}(\boldsymbol{\theta}) = \sum_{i=1}^n \log p_i(\boldsymbol{\theta})$, and the corresponding penalized version gives us the high dimensional pseudo-likelihood criteria as

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \{ -\tilde{\ell}(\boldsymbol{\theta}) + \lambda \sum_{j=1}^J \|\boldsymbol{\beta}_j\|_1 \}. \quad (5)$$

Because each term involves only a J -dimensional orthant probability, the pseudo-likelihood is far cheaper to evaluate than the nJ -dimensional likelihood and yet retains the dependence on $(\rho, \boldsymbol{\Sigma})$ through the block means $\boldsymbol{\mu}_i$ and variances $\boldsymbol{\Omega}_{ii}$. While generating samples for the E-step in MCEM was exponential in both number of observations (n)

and number of alternatives (J), the pseudo-likelihood based approach is just linear in n . The pseudo-likelihood reformulates the problem in terms of observation-level probabilities, replacing the joint likelihood with a sum of node-wise log-probabilities. We next present a modified procedure that adapts Algorithm 1 to this pseudo-likelihood criterion. **Adel:** you mean computational cost? explain it further. In the implementation in next section it seems you are generating n , J -dimensional samples so not sure what you are referring to here. I meant computational cost. I am pushing this into the next section, I feel it makes more sense there.

3.1 Implementation and tuning parameter selection

The optimization problem in (5) can be again tackled using the MCEM procedure. While most steps follow exactly from Algorithm 1, rather than generating samples from the nJ -dimensional truncated multivariate normal, here we generate n , J -dimensional truncated multivariate normal samples. We describe the changes in Algorithm 2. While most steps follow similar from Algorithm 1, the substantive change is the Monte Carlo E-step. Under the full likelihood, the latent vector $\tilde{\mathbf{y}} \in \mathbb{R}^{nJ}$ is sampled from an nJ -dimensional truncated normal over the joint region ($\mathbf{D}(\mathbf{y})\tilde{\mathbf{y}} \leq \mathbf{0}$), implied by all observations, in order to estimate the expectation. In the pseudo-likelihood, since our objective factors into node-wise terms, so we replace this single nJ -dimensional draw by n independent draws of J -dimensional truncated normals, one for each observation $\tilde{\mathbf{y}}_i$ and compute the expectation based on the J -dimensional draws. We describe the changes in Algorithm 2.

As noted earlier in Section 2.2, sampling from an nJ -dimensional truncated Gaussian has computational cost that grows exponentially with nJ . In Algorithm 2, we replace this single joint step by n independent draws from J -dimensional truncated normals. Each of the draw for J -dimensional truncated normals has the computational complexity exponential in J . Moreover, because the n draws are independent across observations, their evaluation scales linearly with n and is readily parallelizable, yielding a substantial computational advantage over the joint nJ -dimensional sampling required by the full likelihood.

Cross Validation. PLEM assumes a known fixed parameter λ to induce sparsity. The choice of λ plays a key role in determining the efficacy of our method for variable selection. We implement cross-validation to select the tuning parameter λ . We split the observations into training set and validation set. Based on the observations in the training set, we create a corresponding network matrix $\mathbf{W}_T$ and similarly create $\mathbf{W}_V$ based on the observation in the validation set. Using PLEM, the parameters $\boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\Sigma}, \boldsymbol{\rho})$ are estimated based on the training set. We then compute the accuracy on the validation set using these estimated parameters $\hat{\boldsymbol{\theta}}$. The tuning parameter is then selected as the λ with the best accuracy in

Algorithm 2 Pseudo-Likelihood based Expectation Maximization

Data : $\mathbf{W}$ - interaction matrix, $\mathbf{X}$ - covariate matrix, $\mathbf{y}$ - response vector

Initialize Set $\beta^{(0)} = \beta_0$, $\Sigma^{(0)} = \Sigma_0$, $\rho^{(0)} = \rho_0$.

for $t = 1, 2, \dots, T$ **do**

MC E-Step: Generate samples of the unknown latent variable $\tilde{\mathbf{Y}}$, using the current value of parameter $\beta^{(t-1)}$, $\Sigma^{(t-1)}$, $\rho^{(t-1)}$:

- Set $\mathbf{H}^{(t)} = \mathbf{I}_{nJ} - \rho^{(t-1)}\tilde{\mathbf{W}}$, $\mu^{(t)} = (\mathbf{H}^{(t)})^{-1}\tilde{\mathbf{X}}\beta^{(t-1)}$ and $\Omega^{(t)} = (\mathbf{H}^{(t)})^{-1}(\mathbf{I}_n \otimes \Sigma^{(t-1)})(\mathbf{H}^{(t)})^{-1}$

for $i = 1, 2, \dots, n$ **do**

Generate K samples $\tilde{\mathbf{y}}_{ik}^{(t)} \sim \text{TMVN}_J(\mu_i^{(t)}, \Omega_{ii}^{(t)}; \mathbf{D}^{(y_i)}\tilde{\mathbf{y}} \leq 0)$ **Adel: What is K ?**

Compute the mean to get $\tilde{\mathbf{y}}_i^{(t)} = \sum_{k=1}^K \tilde{\mathbf{y}}_{ik}^{(t)} / K$

- Stack $\tilde{\mathbf{y}}^{(t)} := (\tilde{\mathbf{y}}_1^{(t)\top}, \dots, \tilde{\mathbf{y}}_n^{(t)\top})^\top$
- Define $\tilde{\mathbf{y}}^{*(t)} := \mathbf{H}^{(t)}\tilde{\mathbf{y}}^{(t)}$, $\tilde{\mathbf{y}}_k^{*(t)} := \mathbf{H}^{(t)}\tilde{\mathbf{y}}_k^{(t)}$

M-Step Same as in Algorithm 1

Output $\beta_1^{(T)}, \beta_2^{(T)}, \dots, \beta_J^{(T)}$, $\Sigma^{(T)}$ and $\rho^{(T)}$

the validation set. **Adel: Does cross validation make sense given that W_V and W_T are different?** Since we are cross validating only to select the sparsity level which depends on β , I think it makes sense even if W_T and W_V are different.

Initialization. Initializations can play a strong role in the convergence of an optimization method. We assume no information of Σ and ρ and initialize them **Adel: You mean initialize?** yes, changed. with the identity matrix and zero respectively. For the regression parameters $\beta_1, \beta_2, \dots, \beta_J$ we want to initialize them with sparse entries. ~~In case of a binomial setting, we use regularized GLM (Friedman et al., 2010) which assumes that each observation is independent (no network effect) and estimates the sparse parameters. Unfortunately, the same does not work for multinomial probit models since the work on high dimensional multinomial probit is limited~~ In the binomial case, classical (unpenalized) Probit estimators are unstable when p grows with n and in our regime $p = o(n)$ this manifests as high variance and poor out-of-sample performance. We therefore use a high-dimensional regularized GLM with a binomial link (Probit model with penalty) (Friedman et al., 2010), which provides a well-posed sparse initializer. In contrast, for the multinomial probit model, high-dimensional methodology is far less developed and there is no off-the-shelf sparse estimator (analogous to the binomial GLM lasso) that we can adopt to initialize our algorithm. **Adel: This is vague. What does it mean that it does not work?** We also

don't give any guarantee whatsoever why the regularized GLM work. Thus to solve this initialization problem we divide the problem into J binomial probits. For every choice j , we only consider the set of observations S_j such that $i \in S_j$ if y_i is either 0 or j . We then use a regularized GLM for binomial probit model on the set S_j to initialize β_j .

4 Theoretical Analysis

In this section, we discuss the theoretical properties of the binomial spatial autoregressive (SAR) Probit model. We prove the consistency of the partial likelihood estimator for the regularized probit model. ~~in the sparse high-dimensional settings~~ Adel: Define the high-dimensional setting, relation of n, p, s , what is diverging?. We work in a sparse high-dimensional regime where the number of covariates p may increase with the sample size n but satisfies $p = o(n)$. The true coefficient vector is sparse with the support size s_0 fixed. In particular, we show that the partial likelihood, derived from the full SAR likelihood, is consistent and achieves the standard high-dimensional convergence rates.

Consider the binomial SAR Probit model where the true parameter set is (ρ, β) . In this setting, $\beta \in \mathbb{R}^p$ is a high-dimensional sparse parameter. The model is defined by

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \beta + \epsilon, \quad \mathbf{z} = \mathbf{1}\{\mathbf{y} > 0\}, \quad (6)$$

where ρ is the auto-correlation parameter, $\mathbf{W} \in \mathbb{R}^{n \times n}$ is the interaction matrix, $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the covariate matrix, and ϵ is standard Gaussian error term.

For a sample of n observations, the full log-likelihood function based on the joint distribution of $\mathbf{y}$ is defined as

$$g_n(\rho, \beta) = \log \int_B \phi_n \left((\mathbf{I}_n - \rho \mathbf{W}) \mathbf{y} - \mathbf{X} \beta \right) d\mathbf{y},$$

where

$$B = B_1 \times B_2 \times \cdots \times B_n, \quad \text{with} \quad B_i = \begin{cases} (0, \infty) & \text{if } z_i = 1, \\ (-\infty, 0) & \text{if } z_i = 0, \end{cases}$$

and ϕ_n is the density function of the n -dimensional standard multivariate Gaussian.

4.1 Pseudo-Likelihood and the Corresponding Loss Function

Although the full likelihood may be more efficient, we provide asymptotic theory for the pseudo likelihood. This choice is motivated by the fact that the asymptotic theory of the dependent Probit model is not well established even in fixed-dimensional settings and that estimation using full likelihood is computationally expensive. Adel: But our PLEM algorithm uses pseudo likelihood right? doesn't this motivate the analysis of Pseudo likelihood?

Yes, it does. But since I gave out an algorithm for the full likelihood as well. I thought of motivating it here as well. Can remove it. Based on model (6), since the error terms ϵ are independent standard Gaussian random variables, we can equivalently express the model as $\mathbf{A}\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \epsilon$, which implies that

$$\mathbf{y} \sim \mathcal{N}(\mathbf{A}^{-1}\mathbf{X}\boldsymbol{\beta}, \mathbf{A}^{-2}),$$

where we define the matrix $\mathbf{A} = \mathbf{I}_n - \rho\mathbf{W}$ for notational convenience. Consequently, marginally, each component y_i is normally distributed with mean given by the i^{th} entry of $\mathbf{A}^{-1}\mathbf{X}\boldsymbol{\beta}$ and a variance equal to the i^{th} diagonal element of $\mathbf{A}^{-2}$.

Based on this marginal distribution, we define the pseudo-likelihood and the corresponding pseudo-loss function ℓ_n as

$$\ell_n(\rho, \boldsymbol{\beta}) = -\frac{1}{n} \sum_{i=1}^n \left\{ z_i \log \Phi \left(\frac{\mathbf{e}_i^T \mathbf{A}^{-1} \mathbf{X} \boldsymbol{\beta}}{\|\mathbf{e}_i^T \mathbf{A}^{-1}\|} \right) + (1 - z_i) \log \left(1 - \Phi \left(\frac{\mathbf{e}_i^T \mathbf{A}^{-1} \mathbf{X} \boldsymbol{\beta}}{\|\mathbf{e}_i^T \mathbf{A}^{-1}\|} \right) \right) \right\},$$

where ℓ_n depends on ρ through the matrix $\mathbf{A}$. Alternatively, this pseudo-loss function can be viewed as the likelihood for a model with no correlation. Specifically, if one considers the problem

$$\tilde{\mathbf{y}} \sim \mathcal{N}(\mathbf{A}^{-1}\mathbf{X}\boldsymbol{\beta}, \text{diag}(\mathbf{A}^{-2})), \quad \tilde{\mathbf{z}} = \mathbf{1}\{\tilde{\mathbf{y}} > 0\},$$

where $\text{diag}(\cdot)$ extracts the diagonal elements of a matrix, the negative likelihood of this problem generates the above loss function.

Based on the pseudo likelihood, we construct a regularized loss function to perform estimation in the high-dimensional regime. Specifically, we define the loss function as

$$f_n(\rho, \boldsymbol{\beta}) = -\frac{1}{n} \ell_n(\rho, \boldsymbol{\beta}) + \lambda \|\boldsymbol{\beta}\|_1, \quad (7)$$

where λ is a tuning parameter controlling the amount of ℓ_1 regularization, and $\|\boldsymbol{\beta}\|_1$ denotes the ℓ_1 norm of the parameter vector $\boldsymbol{\beta}$. The inclusion of the regularization term is essential for promoting sparsity in the estimator.

4.2 Consistency of Pseudo-Likelihood

In order to establish the theoretical guarantees for the estimator, we require several conditions on the underlying model components. These conditions are summarized in the following assumption.

Assumption: The following conditions hold:

- **Interaction Matrix:** The interaction matrix $\mathbf{W}$ has zero diagonal elements and non-negative off-diagonal entries. Moreover, the sum of the off-diagonal elements in each row is strictly less than 1.

- **Interaction Parameter:** The auto-correlation parameter ρ is positive and uniformly bounded away from 1; Adel: Do we allow it to depend on n ? If so we should emphasize it in the model and maybe in notation like ρ_n This is similar to β , so we can write that for any sequence of parameters $\{\beta_i, \rho_i\}_{i=1,2,\dots}$ such that $\rho_i \leq 1 - \epsilon$ and $\|\beta_i\|_\infty \leq C_\beta$ for all i . But I feel it will complicate the writing. Maybe we can mention this somewhere. that is, there exists a constant $\epsilon > 0$ such that $\rho \leq 1 - \epsilon$.
- **Parameter Sparsity:** The parameter vector β satisfies $\|\beta\|_\infty \leq C_\beta$ for some constant C_β . Additionally, the true parameter β_0 is assumed to be sparse, with the number of nonzero entries denoted by s_0 .
- **Covariate Bound:** Every entry of the covariate matrix $\mathbf{X} := \{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n\}$, is uniformly bounded above by a constant; specifically, for all $1 \leq i \leq n$, and j , $x_{ij} \leq C_x$, $\|\mathbf{X}_i\|_\infty \leq C_x$, where C_x is a positive constant. Adel: x_{ij} or $\mathbf{X}_{ij}$ I meant every element of the matrix X , changed it to make it clearer. Adel: Any randomness on $\mathbf{X}$? Not possible, I need X to be strictly bounded, allowing for randomness would mean maximum element of X could be of the order of $\sqrt{\log(np)}$.
- **Design Matrix Regularity:** The n -th largest eigenvalue of matrix $\mathbf{X}^\top \mathbf{X}$ is uniformly bounded away from zero, that is the design matrix satisfies $\liminf_{n \rightarrow \infty} \frac{\mathbf{X}^\top \mathbf{X}}{n} > 0$. $\liminf_{n \rightarrow \infty} \lambda_{\min} \left(\frac{\mathbf{X}^\top \mathbf{X}}{n} \right) > 0$ Adel: What does it mean? Changed. Forgot the minimum eigenvalue operator

Adel: We need to add some justification for these assumptions, especially on the interaction matrix and interaction parameter Under these assumptions, we are ready to state the main theorem regarding the estimation error of our proposed estimator.

Theorem. Under the assumptions above, by choosing the regularization parameter as $\lambda = \delta \sqrt{\log p/n}$ in (7), Adel: Use of δ seems strange, how about λ_0 ? there exists constants c and k Adel: use of k is strange as it implies to be integer. You can use $c_1, c_2, \dots$ The proof needs to be updated accordingly independent of δ, n and p , such that,

$$\mathbb{P} \left(\|\hat{\gamma} - \gamma_0\|_2 \geq c\delta \sqrt{\frac{s_0 \log p}{n}} \right) \leq 4p^{1-k\delta^2},$$

where, $\hat{\gamma} = (\hat{\rho}, \hat{\beta})$ is the estimator obtained by minimizing the regularized pseudo-loss function $f_n(\rho, \beta)$ in (7), and $\gamma_0 = (\rho_0, \beta_0)$ denotes the true model parameter.

Theorem 1. Under the assumptions above, by choosing the regularization parameter as $\lambda = \lambda_0 \sqrt{\log p/n}$ in (7), there exists constants c_1 and c_2 independent of λ_0, n and p , such

that,

$$\mathbb{P} \left(\|\hat{\gamma} - \gamma_0\|_2 \geq c_1 \lambda_0 \sqrt{\frac{s_0 \log p}{n}} \right) \leq 4p^{1-c_2 \lambda_0^2},$$

where, $\hat{\gamma} = (\hat{\rho}, \hat{\beta})$ is the estimator obtained by minimizing the regularized pseudo-loss function $f_n(\rho, \beta)$ in (7), and $\gamma_0 = (\rho_0, \beta_0)$ denotes the true model parameter.

The detailed proofs of the supporting lemmas and Theorem 1 are provided in Section B.

4.3 Sketch of the Proof

To establish the high-probability upper bound on the ℓ_2 estimation error of the regularized estimator $\hat{\gamma} = (\hat{\rho}, \hat{\beta})$, our proof proceeds in three main steps. These steps follow the general structure of high-dimensional regularized M-estimation proofs, adapted to our setting of spatially dependent data and a pseudo-likelihood objective.

We start by exploiting the first-order optimality condition of the regularized estimator $\hat{\gamma}$, which minimizes the objective $f_n(\gamma) := f_n(\rho, \beta)$. Letting $\hat{\gamma} = (\hat{\rho}, \hat{\beta})$, we consider coordinate-wise perturbations of $\hat{\beta}$ and apply convexity arguments to derive a bound on the gradient of the loss function ℓ_n at $\hat{\gamma}$, specifically, we prove that $\|\nabla \ell_n(\hat{\gamma})\|_\infty \leq \lambda$.

The most delicate part of the proof is to show that the loss function, evaluated at the true parameter $\gamma_0 = (\rho_0, \beta_0)$, is uniformly small in every coordinate with high probability.

Lemma 1. *Let $\gamma_0 = (\rho_0, \beta_0)$ denote the true parameters. With $\lambda = \lambda_0 \sqrt{\log p/n}$, under the assumptions of Theorem 1, it holds that*

$$\mathbb{P} \left(\|\nabla \ell_n(\gamma_0)\|_\infty > \frac{\lambda}{2} \right) \leq 4p^{1-c \lambda_0^2},$$

for a constant c independent of λ_0, n and p .

Under independence we could simply apply a Bernstein or Hoeffding bound to the loss function and achieve the high-probability results. But in this situation, this route is blocked for two reasons. First, $z_i = 1\{y_i > 0\}$ are spatially dependent through the SAR structure, so possibly, no coordinate of $\mathbf{z}$ is independent of the others. Second, we maximize only a pseudo-likelihood, not the true log-likelihood; as a result, the score components are neither independent nor true likelihood scores, and the classical consistency results no longer apply.

To overcome these obstacles we rewrite every partial derivative, of the pseudo-likelihood, $\partial \ell_n / \partial \rho$ and $\partial \ell_n / \partial \beta_j$'s, as a linear functional of the binary vector $\mathbf{z}$, with bounded deterministic coefficients $\sum_i (z_i a_i - b_i)$. While this functional has expectation zero, the summands are still dependent. We handle the dependence by constructing a Doob martingale, using the loss function, with the filtration based on the coordinates of $\mathbf{z}$. Since changing a coordinate can alter the partial sum by at most a fixed value, the Azuma–Hoeffding inequality

is applicable. Finally, by applying union bounds on the coordinate-wise tails we achieve the high-probability bound in Lemma 1.

Establishing this inequality is the crux of the entire proof. Once the gradient at the true value is shown to be small, the rest of the argument follows using a lower bound on the Hessian of ℓ_n established based on the design matrix regularity assumption and performing a second-order Taylor expansion around γ_0 . Combining all these pieces yields our final result in Theorem 1.

5 Numerical experiments

We study the performance of PLEM using numerical experiments. We broadly divide the the experiments into two categories, binomial and multinomial. In each category we consider a wide range of regimes by varying (a) number of observations, (b) number of covariates and (c) strength of the network.

Metrics: Since we are performing a variable selection problem in this sparse high-dimensional setting, our main goal is to correctly selecting the signal variables while not selecting the nuisance variables. To judge the performance of PLEM we consider the following two key metrics:

- TPR : True positive rate, the proportion of non zero signals selected by our method
- FPR : False positive rate, the proportion of zero signals selected by our method

These are widely used metrics in the high-dimensional variable selection problem.

Even though variable selection is the main problem we are trying to solve here, the other parameters, specifically the spatial autoregressive parameter ρ can be very insightful. The sign of the autoregressive parameter is especially useful in understanding the effect of the network on the utilities. A positive ρ implies that the network has a positive effect and a consumer makes similar choices as their network. Conversely, a negative autoregressive parameter implies that a consumer avoids a choice that their peers made. This type of understanding of the network effect can be really useful in decision making. Thus, we also use the following metric to evaluate our method.

- ρ Sign Consistency : The proportion of simulations where the estimated ρ and the actual ρ have the same sign

Competitors: In the binomial section, we compare PLEM to a regularized GLM with probit link. The regularized GLM assumes that each observation, i.e. each node on the network is independent. It doesn't take into account the dependence structure of the utilities introduced by the SAR model.

5.1 Binomial Probit: PLEM vs Regularized GLM

For simulations in the binomial category, we need to define a normalized interaction matrix $\mathbf{W}$, the covariate matrix $\mathbf{X}$, the sparse coefficient $\boldsymbol{\beta}$ and the autoregressive parameter ρ . With the help of all of them, we can construct our latent utilities and the corresponding response vector $\mathbf{y}$ as

$$u_i = \rho \sum_{p \neq i}^n \mathbf{W}_{ip} u_p + \mathbf{x}_i^T \boldsymbol{\beta} + \epsilon_i, \quad \epsilon_i \stackrel{i.i.d}{\sim} \mathcal{N}(0, 1) \quad (8)$$

$$y_i = \mathbf{1}(u_i > 0) \quad (9)$$

Set-up 1. Consider an Erdős–Rényi random graph for the network structure. The random graph is such that the probability of any two nodes being connected is 0.2. Based on this random graph we create the normalized network matrix $\mathbf{W}$. We fix the number of covariates $p = 50$ and vary the number of observations n in $\{50, 100, 250, 500\}$ to see how PLEM performs with increasing observations. Each covariate $\mathbf{x}_i$ are generated from an independent p-variate Gaussian distribution with mean zero and covariance matrix $4\mathbf{I}$. The sparse signal $\boldsymbol{\beta}$ has only three non-zero entries, $\boldsymbol{\beta}^T = (3, 2, -2, 0, \dots, 0)$. The auto-correlation parameter ρ , is assumed to be 0.3. The latent utilities and the corresponding response vectors are then generated using (8).

With the generated interaction matrix ($\mathbf{W}$), covariates ($\mathbf{X}$) and the response vector ($\mathbf{y}$), we use PLEM to estimate the parameters. For each (n, p) , we repeat the experiment 100 times. We set k , number of latent variables sampled at each MC E-Step at 10 and let the MCEM run for 50 iterations. Table 1 contains the results from this simulation.

Table 1: Set-up 1: Performance of PLEM compared to regularized Probit model with random graph, with respect to number of observations

n	TPR		FPR	
	MCEM	Regularized GLM	MCEM	Regularized GLM
50	0.920 (± 0.178)	0.977 (± 0.109)	0.067 (± 0.097)	0.100 (± 0.082)
100	0.997 (± 0.033)	1.000 (± 0.000)	0.026 (± 0.061)	0.137 (± 0.101)
250	1.000 (± 0.000)	1.000 (± 0.000)	0.008 (± 0.026)	0.197 (± 0.123)
500	1.000 (± 0.000)	1.000 (± 0.000)	0.003 (± 0.015)	0.214 (± 0.120)

With the results in Table 1, we can see the true positive rate obtained by applying PLEM and regularized GLM. We report the mean TPR and the standard error for both the methods. Both the methods perform extremely well in detecting the signals from the noise and their TPR converge to 1 as n grows.

The second part contains the mean false positive rate and the standard errors for both the methods as n grows. Our method performs extremely well here and the FPR decreases as n increases which is desirable. As n becomes extremely large, the FPR using PLEM for variable selection goes to 0. On the other hand the FPR for regularized GLM is worse and it indeed keeps increasing as n grows.

Set-up 2. The random graph is a simple network structure, but in reality assuming all two nodes being connected with same probability is unrealistic. A better assumption is the existence of communities, where within community the probability of connection is high, while two nodes which do not belong to the same community have low probability of connection. A stochastic block model provides the required framework. In this set-up, we keep all the settings similar to the previous set-up but change the network structure. Rather than a random graph, we use a stochastic block model with three communities. All the nodes are randomly put into the three communities with probability $(0.4, 0.4, 0.2)$. In this SBM, the probability of connection within community is set to 0.2 and that of across communities is 0.01.

Table 2: Set-up 2: Performance of PLEM compared to regularized Probit model with stochastic block model, with respect to number of observations

n	TPR		FPR	
	MCEM	Regularized GLM	MCEM	Regularized GLM
50	0.913 ($\pm$ 0.175)	0.973 ($\pm$ 0.102)	0.077 ($\pm$ 0.117)	0.086 ($\pm$ 0.075)
100	0.987 ($\pm$ 0.066)	1.000 ($\pm$ 0.000)	0.016 ($\pm$ 0.045)	0.134 ($\pm$ 0.093)
250	1.000 ($\pm$ 0.000)	1.000 ($\pm$ 0.000)	0.005 ($\pm$ 0.017)	0.180 ($\pm$ 0.104)
500	1.000 ($\pm$ 0.000)	1.000 ($\pm$ 0.000)	0.003 ($\pm$ 0.010)	0.206 ($\pm$ 0.111)

Table 2 contains the TPR and the FPR results from this simulation. The story is similar to the previous case where in terms of true positive rate both PLEM and regularized GLM grows similarly and converges to 1 as n becomes large. But in terms of the false positive rate, PLEM outperforms regularized GLM. Similar to the previous setting, while using our method the FPR does converge to 0 as n grows, the same is not true for regularized GLM.

Set-up 3. Next, we aim to study the performance of PLEM as the number of covariates p increases. To achieve this we set the number of observations $n = 100$ while the number of covariates p increases in the set $\{50, 100, 200, 500\}$. All the auxiliary parameters remain the same as the previous set-ups and we use a stochastic block model to generate the interaction matrix.

We see the results of this setup in Table 3. In terms of true positive rate with $n = 100$,

Table 3: Set-up 3: Performance of PLEM compared to regularized Probit model with stochastic block model, with respect to number of covariates

p	TPR		FPR	
	MCEM	Regularized GLM	MCEM	Regularized GLM
50	1.000 (± 0.000)	1.000 (± 0.000)	0.008 (± 0.022)	0.166 (± 0.100)
100	1.000 (± 0.000)	1.000 (± 0.000)	0.002 (± 0.008)	0.109 (± 0.068)
250	1.000 (± 0.000)	1.000 (± 0.000)	0.001 (± 0.007)	0.063 (± 0.042)
500	1.000 (± 0.000)	1.000 (± 0.000)	0.000 (± 0.001)	0.037 (± 0.029)

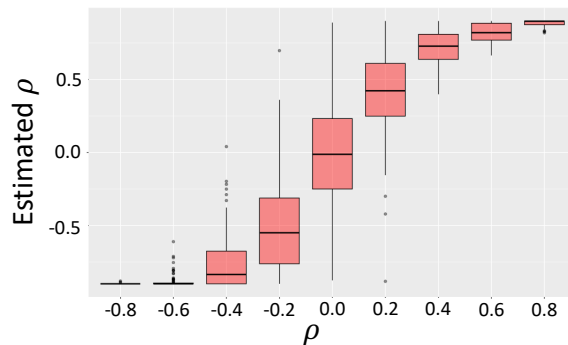


Figure 2: Set-up 4: Binomial setting ρ estimation using PLEM

both the methods perform perfectly and detect all the signals. As p increases, it would be intuitive to assume that the problem becomes harder and both the methods would end up selecting more nuisance variables and hence have a higher FPR. But the results show that as p increases, the FPR decreases using both the methods. This is because even if the absolute number of detected nuisance variables increases, the proportion (FPR) decreases since p increases. Comparing both the methods give us the similar result as previous ones, where PLEM outperforms the regularized GLM.

Set-up 4. In the previous settings we see the ρ sign consistency of our method. To better analyse the estimates of the autoregressive parameter ρ , we conduct this experiment. We set the number of observations $n = 200$ and the number of covariates $p = 50$. We construct the network using the stochastic block model with the same specifications as in the previous set-ups. The covariates are again generated as independent p -variate Gaussian distribution with mean zero and covariance matrix $4\mathbf{I}$ and our sparse signal is $\beta^T = (3, 2, -2, 0, \dots, 0)$ as well. Since the autoregressive parameter $\rho \in [-1, 1]$ we vary it in $\{-0.8, -0.6, -0.4, -0.2, 0, 0.2, 0.4, 0.6, 0.8\}$ and see the ρ estimation. For each ρ in the set, we repeat the experiment 100 times.

For this setup, we create a boxplot in Figure 2. The mean ρ estimation performs well for

all the different values of actual ρ . Observe that as the true ρ moves towards the extremes, near -1 and 1 , the uncertainty of the ρ estimation decreases significantly. This is intuitive since ρ estimation becomes an easier problem as the magnitude of ρ is larger.

5.2 Multinomial Probit

In the multinomial section, we conduct experiments with 3 choices ($J = 2$). For each experiment in addition to the interaction matrix $\mathbf{W}$, covariate matrix $\mathbf{X}$, two sparse coefficient β_1 , β_2 and the autoregressive parameter ρ , we also need to define the matrix Σ , the covariance across choices. In our simulations Σ is a 2×2 matrix since we have $J = 2$. With all the above defined, here we construct the latent utilities and responses based on the SAR structure.

Set-up 5. Similar to the binomial setup, we first see the performance of PLEM with different values of n . The interaction matrix is generated with the stochastic block model where the nodes are randomly put into the three communities with probability $(0.4, 0.4, 0.2)$. And, the probability of connection within community is 0.2 and that of across communities is 0.01. We fix the number of covariates $p = 50$ and increase the number of observations n from 50 to 200. In the multinomial case here (with $J = 2$), we need to two sets of covariates for the two alternatives. The covariates for both the alternatives $\mathbf{x}_1\mathbf{i}$, $\mathbf{x}_2\mathbf{i}$ are generated from independent p-variate Gaussian distribution with mean zero and covariance matrix $4\mathbf{I}$. Corresponding to the two alternatives there are two sparse signals. We assume that both the coefficients have sparsity 3, i.e. there are only three non-zero coefficients in both. The sparse signal $\beta_1^T = (3, 2, -2, 0, \dots, 0)$ and the second sparse signal $\beta_2^T = (2, -2, 3, 0, \dots, 0)$. The auto-correlation parameter ρ , is set to be 0.3. The covariance across the latent utilities Σ has 1 on the diagonals and 0.3 on the off diagonals. The latent utilities and the corresponding response vectors are then generated using (1). Table 4 contains the results from this simulation.

Table 4: Set-up 4: Performance of PLEM compared to regularized Probit model in multinomial setting, with respect to number of observations

n	TPR	FPR
50	0.730 (± 0.183)	0.275 (± 0.132)
100	0.892 (± 0.149)	0.159 (± 0.192)
250	0.998 (± 0.017)	0.019 (± 0.074)
500	1.000 (± 0.000)	0.000 (± 0.000)

Even in the multinomial case with two alternative choices, the TPR grows and converges

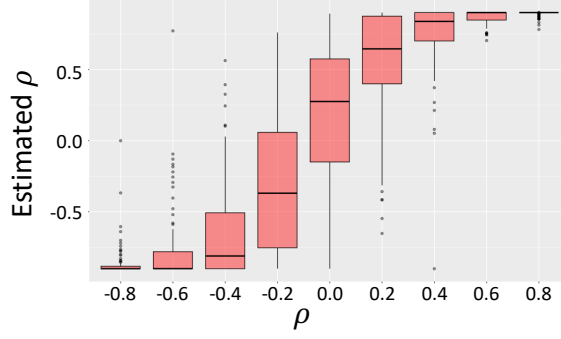


Figure 3: Set-up 7: ρ estimation in a multinomial setting using PLEM.

to 1 and the FPR decreases to 0. Compared to the binomial case, the convergence is slower. This is expected, since with 2 sets of parameters β_1 and β_2 , the effective sample size for estimation is halved.

Set-up 6. Next, we aim to study the performance of PLEM in the multinomial setting, when number of covariates p increases. We set the number of observations $n = 100$ and increase the number of covariates p as $\{50, 100, 200, 500\}$. The covariates, coefficients, autoregressive parameter and the covariance across latent utilities remain the same as the previous set-up and we use a stochastic block model to generate the interaction matrix.

Table 5: Set-up 4: Performance of PLEM compared to regularized Probit model in multinomial setting, with respect to number of covariates

p	TPR	FPR
50	0.987 (± 0.051)	0.021 (± 0.086)
100	0.968 (± 0.088)	0.034 (± 0.097)
250	0.955 (± 0.103)	0.030 (± 0.079)
500	0.937 (± 0.129)	0.034 (± 0.056)

We see the performance of PLEM as p increases. As expected, the performance decreases with increasing p , but we still see reasonable TPR and FPR with larger p 's.

Set-up 7. We also study the performance of our estimates of the autoregressive parameter ρ . Similar to Set-up 4, we vary ρ in $\{-0.8, -0.6, -0.4, -0.2, 0, 0.2, 0.4, 0.6, 0.8\}$ and see the estimation results. Both β_1 and β_2 are set as in the previous set-ups where $\beta_1^T = (3, 2, -2, 0, \dots, 0)$ and $\beta_2^T = (2, -2, 3, 0, \dots, 0)$. The number of observations is set at 200 and the number of covariates 50. For each ρ , we repeat the experiment 100 times.

Compared to the binomial case Figure 2, we can see that the estimates of ρ in the multinomial case in Figure 3 have higher uncertainty.

6 Real data analysis

We utilize a dataset of verified Facebook pages collected via the Facebook Graph API in November 2017 (Rozemberczki et al., 2021), restricted to four categories as defined by Facebook: politicians, government organizations, television shows, and companies. The dataset contains a page-page graph, where the node represents the Facebook page and an undirected edge between two nodes indicates that the pages have mutually “liked” one another. From each page’s short owner-provided description we extract binary features indicating the presence of specific words, and the page’s true category serves as the discrete outcome we wish to predict.

To model these outcomes, we employ the PLEM framework, which combines a probit component on the node-level text features with a spatial autoregressive component on the network of mutual likes. The original dataset comprises 22470 pages and 4714 unique word features. Because both the feature matrix and the page–page graph are extremely sparse, we apply a two-stage subsetting procedure. First, we select only those words that appear in at least 1000 pages, yielding 38 features. Second, we retain only pages that contain at least 7 of these selected words. The resulting data contain 3600 pages and 38 binary features.

In our study, we randomly partition the dataset into a training set of 1000 observations and a test set of 2600 observations, and we report all predictive results on the latter. We benchmark three models: the standard multinomial probit (MNP), the spatial autoregressive multinomial probit (SAR-MNP) without regularization, and our proposed regularized SAR-MNP. To create a high-dimensional, sparse setting, we augment each observation’s feature vector with additional nuisance covariates drawn independently from a Bernoulli distribution. We repeat the experiment 100 times by adding different noise covariates.

In Figure 4, we show the accuracy of the three models with a 90% band with varying levels of noise; no noise, 40 noise covariates, 80 noise covariates, 120 noise covariates, 160 noise covariates and 200 noise covariates. We note the significant performance difference between the regularized SAR MNP in contrast with simple SAR MNP especially as the noise level increases. While SAR MNP utilizes the spatial structure and borrows information to outperform the simpler multinomial probit model, the regularized version maintains performance even when the covariates are extremely noisy.

In Figure 5-6, we plot the one-vs-all and one-vs-one ROC’s for our multiclass prediction with no noise, 120 noise covariates and 200 noise covariates. We can see that the regularized SAR-MNP performs significantly better across all the tasks when there are noise covariates. With no noise however, all the three perform equally well in this setting.

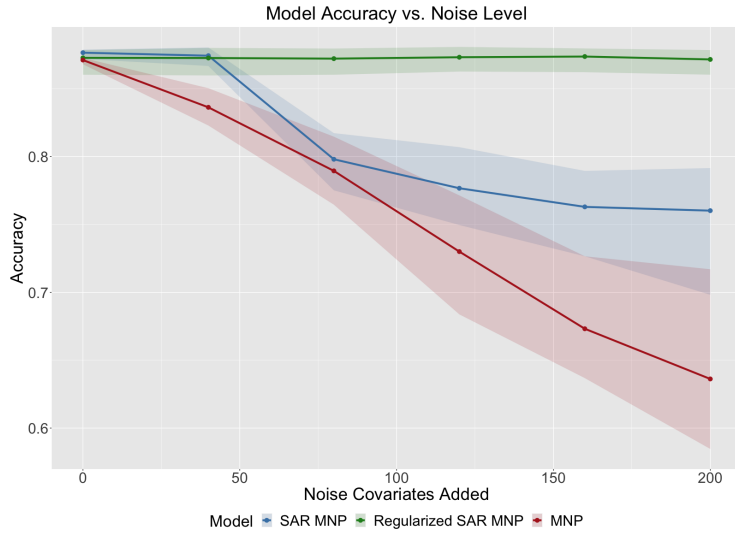


Figure 4: Accuracies of the Multinomial Probit, Spatial Autoregressive Multinomial Probit and PLEM in the testing data with respect to the number of noise covariates

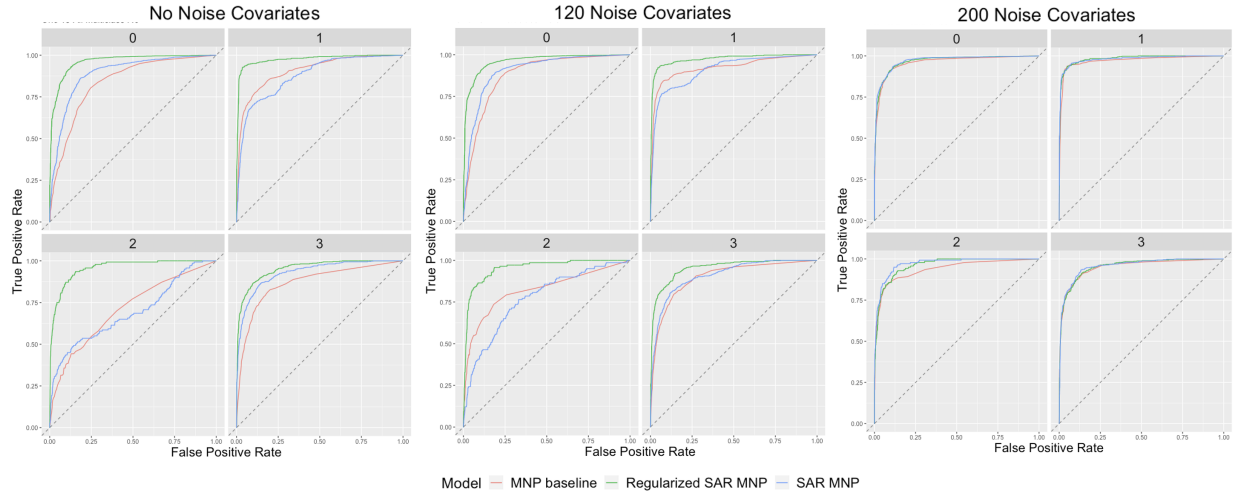


Figure 5: Comparison of MNP, SAR MNP and PLEM in the testing data, One vs All ROC curves, with different noise level

7 Discussion

In this work, we analyze the SAR multinomial probit model in a high dimensional regime. These models are important for discrete choice models in modern datasets with large number of covariates. They also help capturing the correlation across observations through a network which helps in getting better estimates and in turn better prediction performance. The goal was to estimate the sparse high dimensional parameter in such a model. We propose a Pseudo-Likelihood based EM estimation procedure PLEM which is scalable compared to the existing MCEM used as a solution to the SAR probit model.

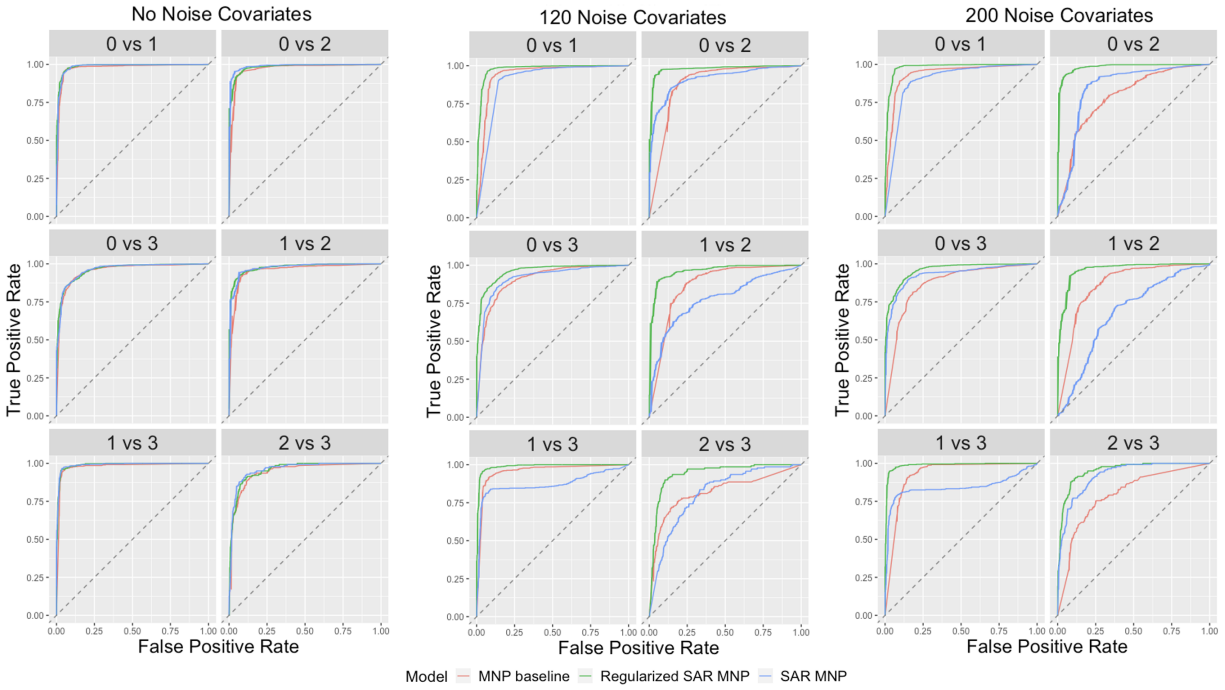


Figure 6: Comparison of MNP, SAR MNP and PLEM in the testing data, One vs One ROC curves, with different noise level

There are several important future directions of our work. In future work, it will be useful to extend PLEM to other spatial structures. ~~The current model only accounts for discrete choices with no natural ordering, but in a wide number of cases, such choices are indeed ordered.~~ Adel: give concrete examples The current model only accounts for discrete choices with no natural ordering, but in many empirical applications, the alternatives are inherently ordered, such as the frequency of ride-hailing usage analyzed in Alemi et al. (2019), the ordinal scale of driver injury severity in traffic crashes studied in Kockelman and Kweon (2002) and physiotherapists' beliefs about pain control in Daykin and Moffatt (2002), all of which work in ordinal setting. It will be interesting to see the applicability of the PLEM in such ordered MNP models.

~~Our current implementation requires us to specify a grid of tuning parameter to select the sparsity level of the parameters and then using cross validation to select the best tuning parameter. In future, we plan to develop an efficient algorithm for the entire regularization path (Efron et al., 2004; Friedman et al., 2010; Tibshirani, 2011), of this problem.~~

Adel: Last sentence does nto read well Our current implementation relies on specifying a grid of tuning parameters to control the sparsity level of the estimates, followed by cross-validation to choose the optimal value. Building on this, we plan to develop an efficient algorithm that can compute the parameters for the entire regularization path (Efron et al., 2004; Friedman et al., 2010; Tibshirani, 2011), enabling us to obtain solutions for all penalty

levels in a single run rather than solving the problem repeatedly for each grid point.

Acknowledgments

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SUPPLEMENTARY MATERIAL

A Details of the M-step

As describe in Section 2.1 the M step of the MCEM estimation for SAR Probit model consists of minimizing the loss computed in the E-step

$$\boldsymbol{\theta}^{(t)} := \min_{\boldsymbol{\theta}} Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(t-1)})$$

. This is a 3-stage process to estimate different set of parameters.

- Stage 1: $\boldsymbol{\beta}^{(t)} := \min_{\boldsymbol{\beta}} Q(\boldsymbol{\beta} | \boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{(t-1)}, \rho = \rho^{(t-1)})$
- Stage 2: $\boldsymbol{\Sigma}^{(t)} := \min_{\boldsymbol{\Sigma}} Q(\boldsymbol{\Sigma} | \boldsymbol{\beta} = \boldsymbol{\beta}^{(t)}, \rho = \rho^{(t-1)})$
- Stage 3: $\rho^{(t)} := \min_{\rho} Q(\rho | \boldsymbol{\beta} = \boldsymbol{\beta}^{(t)}, \boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{(t)})$

Stage 1: Using (2), we know that $l(\boldsymbol{\theta} | \tilde{\mathbf{X}}, \tilde{\mathbf{y}}_k^{(t)}) = \log \phi_{nJ}(\mathbf{H}\tilde{\mathbf{y}}_k^{(t)} | \tilde{\mathbf{X}}\boldsymbol{\beta}, \mathbf{I}_n \otimes \boldsymbol{\Sigma})$. Conditioned on $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{(t-1)}$, we have

$$l(\boldsymbol{\beta} | \tilde{\mathbf{X}}, \tilde{\mathbf{y}}_k, \boldsymbol{\Sigma}^{(t-1)}) = \log \phi_{nJ}\left((\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{-1/2})\mathbf{H}\tilde{\mathbf{y}}_k^{(t)} | (\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{-1/2})\tilde{\mathbf{X}}\boldsymbol{\beta}, \mathbf{I}_{nJ}\right)$$

and

$$\frac{1}{K} \sum_{k=1}^K l(\boldsymbol{\theta} | \tilde{\mathbf{X}}, \tilde{\mathbf{y}}_k^{(t)}, \boldsymbol{\Sigma}^{(t-1)}) = \log \phi_{nJ}\left((\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{-1/2})\mathbf{H}\tilde{\mathbf{y}}^{(t)} | (\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{-1/2})\tilde{\mathbf{X}}\boldsymbol{\beta}, \mathbf{I}_{nJ}\right)$$

where $\tilde{\mathbf{y}}^{(t)} = \sum_{k=1}^K \tilde{\mathbf{y}}_k^{(t)} / K$.

This means that minimizing the conditional expectation boils down to the lasso problem with $\mathbf{Y} = (\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{-1/2})\mathbf{H}\tilde{\mathbf{y}}^{(t)}$ and $\mathbf{X} = (\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{-1/2})\tilde{\mathbf{X}}$. Adel: the connection to next step is missing We explicitly demonstrate this equivalence below.

$$\begin{aligned} \min_{\boldsymbol{\beta}} Q(\boldsymbol{\beta} | \boldsymbol{\theta}^{(t-1)}, \boldsymbol{\Sigma}^{(t-1)}) &= \min_{\boldsymbol{\beta}} \frac{1}{K} \sum_{k=1}^K l(\boldsymbol{\theta} | \tilde{\mathbf{X}}, \tilde{\mathbf{y}}_k^{(t)}, \boldsymbol{\Sigma}^{(t-1)}) + \lambda \|\boldsymbol{\beta}\|_1 \\ &= \min_{\boldsymbol{\beta}} \log \phi_{nJ}\left((\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{-1/2})\mathbf{H}\tilde{\mathbf{y}}^{(t)} | (\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{-1/2})\tilde{\mathbf{X}}\boldsymbol{\beta}, \mathbf{I}_{nJ}\right) + \lambda \|\boldsymbol{\beta}\|_1 \\ &= \min_{\boldsymbol{\beta}} \log \phi_{nJ}\left(\mathbf{Y} | \mathbf{X}\boldsymbol{\beta}, \mathbf{I}_{nJ}\right) + \lambda \|\boldsymbol{\beta}\|_1. \end{aligned}$$

Stage 2: Want to minimize loss conditioned on $\boldsymbol{\beta} = \boldsymbol{\beta}^{(t)}$, again using (2) $l(\boldsymbol{\Sigma} | \tilde{\mathbf{X}}, \tilde{\mathbf{y}}_k^{(t)}, \boldsymbol{\beta}^{(t)}) = \log \phi_{nJ}(\mathbf{H}\tilde{\mathbf{y}}_k^{(t)} | \tilde{\mathbf{X}}\boldsymbol{\beta}^{(t)}, \mathbf{I}_n \otimes \boldsymbol{\Sigma})$ or $\log \phi_{nJ}(\mathbf{H}\tilde{\mathbf{y}}_k^{(t)} - \tilde{\mathbf{X}}\boldsymbol{\beta}^{(t)} | \mathbf{0}, \mathbf{I}_n \otimes \boldsymbol{\Sigma})$.

Define $\tilde{\mathbf{y}}_k^{*(t)} := \mathbf{H}\tilde{\mathbf{y}}_k^{(t)}$, and $\epsilon_{jik}^{(t)} := \tilde{y}_{kji}^{*(t)} - \mathbf{x}_{ji}\boldsymbol{\beta}_{ji}^{(t)}$. Since the covariance matrix of $\mathbf{H}\tilde{\mathbf{y}}_k^{(t)} - \tilde{\mathbf{X}}\boldsymbol{\beta}^{(t)}$ is $\mathbf{I}_n \otimes \boldsymbol{\Sigma}$

$$\tilde{\boldsymbol{\epsilon}}_{ik}^{(t)} = \begin{pmatrix} \epsilon_{1ik}^{(t)} \\ \epsilon_{2ik}^{(t)} \\ \vdots \\ \epsilon_{Jik}^{(t)} \end{pmatrix} \stackrel{i.i.d}{\sim} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad \forall i, k.$$

Based on the distribution of $\tilde{\boldsymbol{\epsilon}}_{ik}^{(t)}$, the structure of $\boldsymbol{\Sigma} = \left[\begin{array}{c|c} 1 & \boldsymbol{\eta}^T \\ \hline \boldsymbol{\eta} & \boldsymbol{\Phi} + \boldsymbol{\eta}\boldsymbol{\eta}^T \end{array} \right]$ and the fact that the penalty does not involve $\boldsymbol{\Sigma}$ we get the estimate of $\boldsymbol{\Sigma}^{(t)}$.

Stage 3: Redefining (2), it is easy to see that $(\mathbf{I}_n \otimes \boldsymbol{\Sigma})^{-1/2}(\mathbf{H}\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\boldsymbol{\beta}) \sim \mathcal{N}(0, \mathbf{I}_n)$. Using this, conditional on $\boldsymbol{\beta} = \boldsymbol{\beta}^{(t)}$ and $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^{(t)}$ we need to minimize,

$$\sum_{k=1}^K \|(\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{(t)})^{-1/2}(\mathbf{H}\tilde{\mathbf{y}}_k - \tilde{\mathbf{X}}\boldsymbol{\beta}^{(t)})\|^2,$$

where $\mathbf{H} = \mathbf{I}_{nJ} - \rho\tilde{\mathbf{W}}$ and we are trying to minimize with respect to ρ . Minimizing this gives us

$$\rho^{(t)} = \frac{\sum_{k=1}^K \left(\tilde{\mathbf{y}}_k^{(t)} \right)^T (\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{-1}) \tilde{\mathbf{W}} \tilde{\mathbf{y}}_k^{(t)} - K \left((\tilde{\mathbf{X}}\boldsymbol{\beta}^{(t)})^T (\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{-1}) \tilde{\mathbf{W}} \tilde{\mathbf{y}}^{(t)} \right)}{\sum_{k=1}^K \left(\tilde{\mathbf{y}}_k^{(t)} \right)^T \tilde{\mathbf{W}}^T (\mathbf{I}_n \otimes \boldsymbol{\Sigma}^{-1}) \tilde{\mathbf{W}} \tilde{\mathbf{y}}_k^{(t)}}.$$

Since the penalty does not contain ρ , this is indeed the estimate in this iteration.

B Proof of Theorem 1

We start by establishing that the gradient of ℓ_n is bounded at the minimizer of the regularized loss function f_n . In particular, we will show that $\|\nabla \ell_n(\hat{\boldsymbol{\gamma}})\|_\infty \leq \lambda$, where $\hat{\boldsymbol{\gamma}}$ is defined as the solution to [Adel: red what?](#) $f_n(\boldsymbol{\gamma})$, $\boldsymbol{\gamma} := (\rho, \boldsymbol{\beta})$, where $f_n(\boldsymbol{\gamma})$ is given by

$$f_n(\boldsymbol{\gamma}) = \ell_n(\boldsymbol{\gamma}) + \lambda \|\boldsymbol{\beta}\|_1,$$

and the function ℓ_n is given by

$$\ell_n(\boldsymbol{\gamma}) = -\frac{1}{n} \sum_{i=1}^n \left\{ z_i \log \Phi \left(\frac{\mathbf{e}_i^T \mathbf{A}^{-1} \mathbf{X} \boldsymbol{\beta}}{\|\mathbf{e}_i^T \mathbf{A}^{-1}\|} \right) + (1 - z_i) \log \left(1 - \Phi \left(\frac{\mathbf{e}_i^T \mathbf{A}^{-1} \mathbf{X} \boldsymbol{\beta}}{\|\mathbf{e}_i^T \mathbf{A}^{-1}\|} \right) \right) \right\},$$

where $\boldsymbol{\gamma} := (\rho, \boldsymbol{\beta})$, and the matrix $\mathbf{A} = \mathbf{I}_n - \rho \mathbf{W}$ depends on the auto-correlation parameter ρ .

To establish the gradient bound, fix an index $1 \leq i \leq p$ and define the univariate function

$$\ell_n^{(i)}(b) = \ell_n(\hat{\rho}, \hat{\beta}_1, \dots, \hat{\beta}_{i-1}, b, \hat{\beta}_{i+1}, \dots, \hat{\beta}_p),$$

which is obtained by keeping all components of β fixed at $\hat{\beta}$ except the i th component, which is allowed to vary as b .

By the definition of $\hat{\gamma}$ as the minimizer of $f_n(\gamma) = \ell_n(\gamma) + \lambda \|\beta\|_1$, we have for any $b \in \mathbb{R}$,

$$\ell_n^{(i)}(\hat{\beta}_i) + \lambda |\hat{\beta}_i| \leq \ell_n^{(i)}(b) + \lambda |b|.$$

This inequality implies that for any $b \neq \hat{\beta}_i$, the function $\ell_n^{(i)}(b)$ satisfies Adel: Why?

$$-\lambda \leq \frac{\ell_n^{(i)}(b) - \ell_n^{(i)}(\hat{\beta}_i)}{b - \hat{\beta}_i} \leq \lambda.$$

This inequality implies that for each i (i.e., for each component of β), the corresponding partial derivative of ℓ_n at $\hat{\gamma}$ is bounded in absolute value by λ ,

$$\left| \frac{\partial}{\partial \beta_i} \ell_n(\gamma) \Big|_{\gamma=\hat{\gamma}} \right| \leq \lambda.$$

Adel: to show above, you can just say by optimality $\nabla \ell_n(\hat{\gamma}) + \lambda \partial \|\hat{\beta}\|_1 = 0$, with ∂ the subgradient. Subgradient of l_1 norm is in $[-1, 1]$, so $\|\nabla \ell_n(\hat{\gamma})\|_\infty \leq \lambda$.

Next, we handle the derivative with respect to the parameter ρ . Define

$$\ell_n^{(0)}(p) = \ell_n(p, \hat{\beta}_1, \dots, \hat{\beta}_p).$$

Using similar reasoning as before, since $\hat{\rho}$ minimizes $f_n(\gamma)$ when the β components are fixed, we have $\ell_n^{(0)}(\hat{\rho}) \leq \ell_n^{(0)}(p)$ for any p . This optimality condition forces the derivative

$$\frac{\partial}{\partial \rho} \ell_n(\gamma) \Big|_{\gamma=\hat{\gamma}} = 0.$$

Thus, combining these results, we conclude that $\|\nabla \ell_n(\hat{\gamma})\|_\infty \leq \lambda$.

To establish $\|\nabla \ell_n(\hat{\gamma})\|_\infty \leq \lambda$, note that $\hat{\gamma}$ minimizes $f_n(\gamma) = \ell_n(\gamma) + \lambda \|\beta\|_1$. The first-order optimality condition for this problem implies that $\nabla \ell_n(\hat{\gamma}) + \lambda \mathbf{z} = 0$, where $\mathbf{z} \in \partial \|\hat{\beta}\|_1$ is a subgradient of the ℓ_1 -norm evaluated at $\hat{\beta}$. Since $\mathbf{z}$ is the subgradient of the ℓ_1 norm, each of its components lie in the interval $[-1, 1]$, which yields $\|\nabla \ell_n(\hat{\gamma})\|_\infty = \lambda \|\mathbf{z}\|_\infty \leq \lambda$.

We now proceed to show that the infinity norm of the gradient of ℓ_n evaluated at the true parameter γ_0 is also bounded with high probability. In particular, we have the following lemma.

Lemma 2. Let $\gamma_0 = (\rho_0, \beta_0)$ denote the true parameters. *Adel: γ_0 should be bold throughout*
Changed With $\lambda = \lambda_0 \sqrt{\log p/n}$, under the assumptions of Theorem 1, it holds that

$$\mathbb{P}\left(\|\nabla \ell_n(\gamma_0)\|_\infty > \frac{\lambda}{2}\right) \leq 4p^{1-c\lambda_0^2},$$

for a constant c independent of δ, n and p .

Lemma 2 allows us to define the event

$$\mathcal{G}_n = \left\{ \omega : \|\nabla \ell_n(\gamma_0)\|_\infty \leq \frac{\lambda}{2} \right\}.$$

For the rest of the proof, we assume that the observed data ω lies in $\mathcal{G}_n$. That is, we work on the event where $\|\nabla \ell_n(\gamma_0)\|_\infty \leq \lambda/2$. By the definition of $\hat{\gamma}$ as a minimizer of $f_n(\gamma) = \ell_n(\gamma) + \lambda\|\beta\|_1$, we have

$$\ell_n(\hat{\gamma}) + \lambda\|\hat{\beta}\|_1 \leq \ell_n(\gamma_0) + \lambda\|\beta_0\|_1.$$

Rearranging, this yields

$$\ell_n(\hat{\gamma}) - \ell_n(\gamma_0) \leq \lambda\left(\|\beta_0\|_1 - \|\hat{\beta}\|_1\right). \quad (10)$$

We now use a Taylor series expansion of $\ell_n(\hat{\gamma})$ around γ_0 . By Taylor's theorem, there exists some $\bar{\gamma}$ between $\hat{\gamma}$ and γ_0 such that

$$\ell_n(\hat{\gamma}) - \ell_n(\gamma_0) = \nabla \ell_n(\gamma_0)^T (\hat{\gamma} - \gamma_0) + \frac{1}{2} (\hat{\gamma} - \gamma_0)^T \nabla^2 \ell_n(\bar{\gamma}) (\hat{\gamma} - \gamma_0).$$

Since on $\mathcal{G}_n$ we have $\|\nabla \ell_n(\gamma_0)\|_\infty \leq \lambda/2$, it follows that

$$\nabla \ell_n(\gamma_0)^T (\hat{\gamma} - \gamma_0) \geq -\|\nabla \ell_n(\gamma_0)\|_\infty \|\hat{\gamma} - \gamma_0\|_1 \geq -\frac{\lambda}{2} \|\hat{\gamma} - \gamma_0\|_1.$$

Furthermore, note that the Hessian of the loss function with respect to β , $\partial^2 \ell_n(\gamma)/\partial \beta^2 \geq c_2 X^\top X/n$ for constant c_2 which depends on C_X and C_β . Using the design matrix regularity assumption of $\liminf_{n \rightarrow \infty} \frac{X^\top X}{n} \geq 0$ *Adel: as said before, this doesn't have meaning*, there exists a constant c_H such that the Hessian is uniformly bounded below as $\partial^2 \ell_n(\gamma)/\partial \beta^2 \succeq c_H \mathbf{I}$ in the sense that $(\hat{\gamma} - \gamma_0)^T \nabla^2 \ell_n(\bar{\gamma}) (\hat{\gamma} - \gamma_0) \geq c_H \|\hat{\gamma} - \gamma_0\|_2^2$.

Furthermore, consider the Hessian of the loss function with respect to β , denoted by $\partial^2 \ell_n(\gamma)/\partial \beta^2$. Under the model assumptions, $\partial^2 \ell_n(\gamma)/\partial \beta^2 \succeq c_\delta \frac{X^\top X}{n}$, for some positive constant c_δ that depends on C_x and C_β . The regularity condition on the design matrix,

$$\liminf_{n \rightarrow \infty} \lambda_{\min} \left(\frac{X^\top X}{n} \right) > 0,$$

ensures that there exists a constant $c_H > 0$ such that the Hessian is uniformly bounded below as $\partial^2 \ell_n(\gamma)/\partial \beta^2 \succeq c_H \mathbf{I}$. Consequently, for any point $\bar{\gamma}$ between $\hat{\gamma}$ and the true parameter γ_0 , it holds that $(\hat{\gamma} - \gamma_0)^T \nabla^2 \ell_n(\bar{\gamma})(\hat{\gamma} - \gamma_0) \geq c_H \|\hat{\gamma} - \gamma_0\|_2^2$.

Then we have

$$\ell_n(\hat{\gamma}) - \ell_n(\gamma_0) \geq -\frac{\lambda}{2} \|\hat{\gamma} - \gamma_0\|_1 + \frac{c_H}{2} \|\hat{\gamma} - \gamma_0\|_2^2. \quad (11)$$

Next, let $\mathcal{S} = \{i : \beta_{0,i} \neq 0\}$ denote the support of β_0 . Also, the notation $(\beta)_{\mathcal{S}}$ denotes the subvector of β whose components are indexed by the elements in $\mathcal{S}$, that is, $(\beta)_{\mathcal{S}} = (\beta_i)_{i \in \mathcal{S}}$. We now compare the ℓ_1 -norms of $\hat{\beta}$ and β_0 . Notice that

$$\|\hat{\beta}\|_1 = \|(\hat{\beta})_{\mathcal{S}}\|_1 + \|(\hat{\beta})_{\mathcal{S}^c}\|_1 = \|(\beta_0)_{\mathcal{S}} + (\hat{\beta} - \beta_0)_{\mathcal{S}}\|_1 + \|(\hat{\beta})_{\mathcal{S}^c}\|_1.$$

Since $(\beta_0)_{\mathcal{S}^c} = 0$, using the triangle inequality, it follows from the above equation that

$$\|\hat{\beta}\|_1 \geq \|(\beta_0)_{\mathcal{S}}\|_1 - \|(\hat{\beta} - \beta_0)_{\mathcal{S}}\|_1 + \|(\hat{\beta} - \beta_0)_{\mathcal{S}^c}\|_1.$$

Rearranging and using $\|\beta_0\|_1 = \|(\beta_0)_{\mathcal{S}}\|_1$, we obtain

$$\|(\hat{\beta} - \beta_0)_{\mathcal{S}}\|_1 - \|(\hat{\beta} - \beta_0)_{\mathcal{S}^c}\|_1 \geq \|\beta_0\|_1 - \|\hat{\beta}\|_1. \quad (12)$$

Combining (12) with the earlier bound on the difference $\ell_n(\hat{\gamma}) - \ell_n(\gamma_0)$, (11) and (10), we get

$$\lambda \|(\hat{\beta} - \beta_0)_{\mathcal{S}}\|_1 - \lambda \|(\hat{\beta} - \beta_0)_{\mathcal{S}^c}\|_1 \geq -\frac{\lambda}{2} \|\hat{\gamma} - \gamma_0\|_1 + \frac{c_H}{2} \|\hat{\gamma} - \gamma_0\|_2^2.$$

Next, note that

$$\|\hat{\gamma} - \gamma_0\|_1 = \|(\hat{\beta} - \beta_0)_{\mathcal{S}}\|_1 + \|(\hat{\beta} - \beta_0)_{\mathcal{S}^c}\|_1 + |\hat{\rho} - \rho_0|.$$

Thus, the previous inequality can be rearranged to yield

$$\begin{aligned} \|\hat{\gamma} - \gamma_0\|_2^2 &\leq \frac{\lambda}{c_H} \left\{ 2\|(\hat{\beta} - \beta_0)_{\mathcal{S}}\|_1 - 2\|(\hat{\beta} - \beta_0)_{\mathcal{S}^c}\|_1 + \|\hat{\gamma} - \gamma_0\|_1 \right\} \\ &= \frac{\lambda}{c_H} \left\{ 3\|(\hat{\beta} - \beta_0)_{\mathcal{S}}\|_1 - \|(\hat{\beta} - \beta_0)_{\mathcal{S}^c}\|_1 + |\hat{\rho} - \rho_0| \right\} \\ &\leq \frac{3\lambda}{c_H} \{ \|(\hat{\beta} - \beta_0)_{\mathcal{S}}\|_1 + |\hat{\rho} - \rho_0| \}. \end{aligned}$$

Next, applying the Cauchy-Schwarz inequality, we obtain

$$\|(\hat{\beta} - \beta_0)_{\mathcal{S}}\|_1 + |\hat{\rho} - \rho_0| \leq \sqrt{|\mathcal{S}| + 1} \|\hat{\gamma} - \gamma_0\|_2.$$

Substituting this bound into the inequality for $\|\hat{\gamma} - \gamma_0\|_2^2$, we deduce that

$$\|\hat{\gamma} - \gamma_0\|_2^2 \leq \frac{3\lambda\sqrt{|\mathcal{S}| + 1}}{c_H} \|\hat{\gamma} - \gamma_0\|_2 \leq \|\hat{\gamma} - \gamma_0\|_2 c\lambda\sqrt{|\mathcal{S}|},$$

for some constant c . **Adel: Why c_H is changed to k ? Changed to c_H .** Note that c does not depend on n, p or λ_0 .

Finally, recalling that the regularization parameter is chosen as $\lambda = \lambda_0 \sqrt{\frac{\log(p)}{n}}$, we conclude that under $\mathcal{G}_n$, with probability $1 - 4p^{1-k\lambda_0^2}$,

$$\|\hat{\gamma} - \gamma_0\|_2 \leq c\lambda_0 \sqrt{\frac{|\mathcal{S}| \log p}{n}}.$$

This completes the proof.

B.1 Proof of Lemma 2

The pseudo-likelihood, ℓ_n , is defined as

$$\ell_n(\rho, \beta) = -\frac{1}{n} \sum_{i=1}^n \left\{ z_i \log \Phi \left(\frac{\mathbf{e}_i^T \mathbf{A}^{-1} \mathbf{X} \beta}{\|\mathbf{e}_i^T \mathbf{A}^{-1}\|} \right) + (1 - z_i) \log \left(1 - \Phi \left(\frac{\mathbf{e}_i^T \mathbf{A}^{-1} \mathbf{X} \beta}{\|\mathbf{e}_i^T \mathbf{A}^{-1}\|} \right) \right) \right\}, \quad (13)$$

where the matrix $\mathbf{A}$ is given by $\mathbf{A} = \mathbf{I} - \rho \mathbf{W}$.

The gradient of ℓ_n is defined as

$$\nabla \ell_n(\rho, \beta) = \left(\frac{\partial \ell_n}{\partial \rho}, \frac{\partial \ell_n}{\partial \beta_1}, \frac{\partial \ell_n}{\partial \beta_2}, \dots, \frac{\partial \ell_n}{\partial \beta_p} \right).$$

The first step in our analysis is to show that the partial derivatives of ℓ_n are well-concentrated. A detailed proof of these concentration results is provided in Section B.2.

Lemma 3. *Under the assumptions of Theorem 1, for every $t > 0$, the following bounds hold for ℓ_n as defined above in (13):*

$$\mathbb{P} \left(\left| \frac{\partial \ell_n}{\partial \rho} \Big|_{\gamma_0} \right| \geq t \right) \leq 2e^{-c_\rho n t^2} \quad \text{and} \quad \mathbb{P} \left(\left| \frac{\partial \ell_n}{\partial \beta_j} \Big|_{\gamma_0} \right| \geq t \right) \leq 2e^{-c_\beta n t^2} \quad \forall j \in \{1, 2, \dots, p\},$$

where $\gamma_0 = (\rho_0, \beta_0)$ denotes the true parameters, and c_ρ and c_β are positive constants.

Using these concentration inequalities, we can apply a union bound over all $p + 1$ coordinates to control the maximum deviation of the gradient. In particular, we have

$$\mathbb{P} \left(\|\nabla \ell_n(\gamma_0)\|_\infty \geq t \right) \leq 2(p + 1)e^{-4c n t^2} \leq 4p e^{-4c n t^2},$$

by choosing the positive constant $c = \min\{c_\rho, c_\beta\}/4$.

Now, by choosing $t = \lambda/2$, where $\lambda = \lambda_0 \sqrt{\log p/n}$, we obtain the final bound for the partial derivatives at the true parameter,

$$\mathbb{P} \left(\|\nabla \ell_n(\gamma_0)\|_\infty \geq \frac{\lambda}{2} \right) \leq 4p \exp \left(-4c n \left(\frac{\lambda}{2} \right)^2 \right) = 4p \exp \left(-c \lambda_0^2 \log p \right) = 4p^{1-c\lambda_0^2}.$$

B.2 Proof of Lemma 3

We have defined the pseudo-likelihood, ℓ_n , as

$$\ell_n(\rho, \beta) = -\frac{1}{n} \sum_{i=1}^n \left\{ z_i \log \Phi \left(\frac{\mathbf{e}_i^T \mathbf{A}^{-1} \mathbf{X} \beta}{\|\mathbf{e}_i^T \mathbf{A}^{-1}\|} \right) + (1 - z_i) \log \left(1 - \Phi \left(\frac{\mathbf{e}_i^T \mathbf{A}^{-1} \mathbf{X} \beta}{\|\mathbf{e}_i^T \mathbf{A}^{-1}\|} \right) \right) \right\},$$

where $\mathbf{A} = \mathbf{I} - \rho \mathbf{W}$.

Our first step is to simplify the partial derivatives $\partial \ell_n / \partial \rho|_{\gamma_0}$ and $\partial \ell_n / \partial \beta_j|_{\gamma_0}$ with the following lemma.

Lemma. *There exist bounded constants a_{ij} and b_{ij} for $1 \leq i \leq n$ and $0 \leq j \leq p$ (with $j = 0$ corresponding to ρ and $j \geq 1$ corresponding to β_j that depend only on γ_0 , $\mathbf{W}$, and $\mathbf{X}$, such that the partial derivatives can be written as*

$$\frac{\partial \ell_n}{\partial \rho} \Big|_{\gamma_0} = -\frac{1}{n} \sum_{i=1}^n \left(z_i a_{i0} - b_{i0} \right)$$

and for every $j \in \{1, 2, \dots, p\}$,

$$\frac{\partial \ell_n}{\partial \beta_j} \Big|_{\gamma_0} = -\frac{1}{n} \sum_{i=1}^n \left(z_i a_{ij} - b_{ij} \right).$$

Lemma 4. *There exist constants a_{ij} and b_{ij} for $1 \leq i \leq n$ and $0 \leq j \leq p$ (with $j = 0$ corresponding to ρ and $j \geq 1$ corresponding to β_j that depend only on γ_0 , $\mathbf{W}$, and $\mathbf{X}$, such that the partial derivatives can be written as*

$$\frac{\partial \ell_n}{\partial \rho} \Big|_{\gamma_0} = -\frac{1}{n} \sum_{i=1}^n \left(z_i a_{i0} - b_{i0} \right)$$

and for every $j \in \{1, 2, \dots, p\}$,

$$\frac{\partial \ell_n}{\partial \beta_j} \Big|_{\gamma_0} = -\frac{1}{n} \sum_{i=1}^n \left(z_i a_{ij} - b_{ij} \right).$$

The proof of Lemma 4 is provided in Section B.4. Further, note that the pseudo log-likelihood is the sum of likelihood for each observation. Since the expectation of partial derivatives of log-likelihood is zero, with the linearity of expectation, the partial derivatives of the pseudo-likelihood is mean zero as well, $\mathbb{E}[\partial \ell_n / \partial \rho|_{\gamma_0}] = 0$ and $\mathbb{E}[\partial \ell_n / \partial \beta_j|_{\gamma_0}] = 0$.

The key step towards proving the concentration inequality for all the partial derivatives is the following lemma which exploits the structure of the partial derivatives in the above lemma.

Lemma 5. Let $\mathbf{y} \sim \mathcal{N}(\mathbf{A}_0^{-1} \mathbf{X} \boldsymbol{\beta}_0, \mathbf{A}_0^{-2})$ and define $\mathbf{z} = \mathbf{1}\{\mathbf{y} > 0\}$. Consider a function f defined on $\mathbf{z} = (z_1, z_2, \dots, z_n)$ by

$$f(\mathbf{z}) = n^{-1} \sum_{i=1}^n (z_i a_i - b_i),$$

where the constants a_i and b_i satisfy $\sup_i a_i \leq a$ and $\sup_i b_i \leq b$ for some global constants a and b , and assume that $\mathbb{E}[f(\mathbf{z})] = 0$. Then, for any $t \geq 0$,

$$\mathbb{P}(|f(\mathbf{z})| \geq t) \leq 2e^{-knt^2},$$

for some constant $k > 0$.

The proof of Lemma 5 is provided in Section B.3. With this result in hand, note that the partial derivative with respect to ρ can be written as $\partial \ell_n / \partial \rho|_{\gamma_0} = -n^{-1} f(\mathbf{z})$ and thus, for any $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{\partial \ell_n}{\partial \rho}\right|_{\gamma_0} \geq \varepsilon\right) = \mathbb{P}\left(\left|\frac{1}{n} f(\mathbf{z})\right| \geq \varepsilon\right) \leq 2 \exp(-k_\rho n \varepsilon^2).$$

Similarly, for each $j \in \{1, 2, \dots, p\}$, the partial derivative with respect to β_j satisfies

$$\mathbb{P}\left(\left|\frac{\partial \ell_n}{\partial \beta_j}\right|_{\gamma_0} \geq \varepsilon\right) = \mathbb{P}\left(\left|\frac{1}{n} f(\mathbf{z})\right| \geq \varepsilon\right) \leq 2 \exp(-k_j n \varepsilon^2).$$

By defining $k_\beta := \min\{k_1, k_2, \dots, k_p\}$, the concentration inequalities for all the partial derivatives are established. **Adel: I think your a and b in Lemma 5 can depend on s and so p . Taking the minimum of p of k_i 's could be tricky. a and b only depend on s and not p . Assumptions: $p = o(n)$, $\|\boldsymbol{\beta}\|_0 \leq s_0$ and $\|\boldsymbol{\beta}\|_\infty \leq C_\beta$ or $\|\boldsymbol{\beta}\|_1 \leq c$. Just checked the proof, k only depends on $a, \mathbb{P}(y_{j+1} \geq 0 | z_{1:j})$, so it should be independent of p , will update this accordingly.**

B.3 Proof of Lemma 5

Lemma 6. Let $\mathbf{W}$ be the interaction matrix such that $\mathbf{W}$ is symmetric with non-negative off-diagonal entries, zero diagonal, and $\|\mathbf{W}\|_\infty \leq 1$. ρ is the interaction parameter with $|\rho| < 1 - \epsilon$ and let $\mathbf{A} = \mathbf{I} - \rho \mathbf{W}$. Then the following properties hold,

1. $\|\mathbf{A}^{-1}\|_\infty \leq (1 - \rho)^{-1}$, and consequently, $\|\mathbf{A}^{-2}\|_\infty \leq (1 - \rho)^{-2}$.
2. $(\mathbf{A}^{-2})_{ii} \geq 1$, for every $i = 1, \dots, n$.

We begin by constructing a Doob martingale based on $f(\mathbf{z})$. First, we note that since $\sup_i a_i \leq a$ and $\sup_i b_i \leq b$, we have

$$\begin{aligned}\mathbb{E}[|f(\mathbf{z})|] &\leq \mathbb{E}\left[\sum_{i=1}^n |z_i a_i + b_i|\right] \leq \sum_{i=1}^n \mathbb{E}[|z_i a_i| + |b_i|] \\ &\leq \sum_{i=1}^n \mathbb{E}[|a_i| + |b_i|] \leq n(a + b) \leq \infty \quad \text{Adel : ???}\end{aligned}$$

where we use the fact that $z_i \in [0, 1]$ and consequently $\mathbb{E}[|z_i|] \leq 1$. Adel: I think you should normalize by $1/n$ and define the U'_i 's accordingly. This way above you have $[|f(\mathbf{z})|/n] \leq a + b$

Next, we define the sequence of random variables:

$$\begin{aligned}U_0 &= \mathbb{E}[f(\mathbf{z})], \\ U_1 &= \mathbb{E}[f(\mathbf{z})|z_1], \\ U_2 &= \mathbb{E}[f(\mathbf{z})|z_{1:2}], \\ &\vdots \\ U_i &= \mathbb{E}[f(\mathbf{z})|z_{1:i}], \\ &\vdots \\ U_n &= \mathbb{E}[f(\mathbf{z})|z_{1:n}],\end{aligned}$$

where $z_{i:j} := \{z_i, z_{i+1}, \dots, z_j\}$. Since $\mathbb{E}[|f(\mathbf{z})|] < \infty$, the sequence $\{U_0, U_1, \dots, U_n\}$ forms a Doob martingale.

The crux of the proof lies in using the Azuma-Hoeffding inequality: Adel: the format here is messed up

Result (Azuma-Hoeffding). *If $\{X_k : k = 0, 1, 2, \dots\}$ is a martingale and there exist constants c_k such that $|X_k - X_{k-1}| \leq c_k$ for all k , then*

$$\mathbb{P}(|X_N - X_0| \geq t) \leq 2 \exp\left(\frac{-t^2}{2 \sum_{k=1}^N c_k^2}\right).$$

To use this inequality, we will show that there exist constants $c_1, c_2, \dots, c_n$ such that $|U_i - U_{i-1}| \leq c_i$ and

$$\sum_{i=1}^n c_i^2 = \frac{n}{2k}$$

for some constant k .

Applying the Azuma-Hoeffding inequality will then yield **Adel: without normalization, below you get $2 \exp(-kt^2/n)$**

$$\begin{aligned}\mathbb{P}(|U_n - U_0| \geq t) &\leq 2 \exp\left(\frac{-t^2}{2(n/2k)}\right) \\ &\implies \mathbb{P}(|f(\mathbf{z})| \geq t) \leq 2 \exp(-knt^2),\end{aligned}$$

where we used the facts that $U_n = f(\mathbf{z})$ and $U_0 = \mathbb{E}[f(\mathbf{z})] = 0$. This would complete the proof.

We begin by constructing a Doob martingale based on $f(\mathbf{z})$. First, we note that since $\sup_i a_i \leq a$ and $\sup_i b_i \leq b$, we have

$$\begin{aligned}\mathbb{E}[|f(\mathbf{z})|] &\leq \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n |z_i a_i + b_i|\right] \leq \frac{1}{n} \sum_{i=1}^n \mathbb{E}[|z_i a_i| + |b_i|] \\ &\leq \frac{1}{n} \sum_{i=1}^n \sum_{i=1}^n \mathbb{E}[|a_i| + |b_i|] \leq (a + b) \leq \infty\end{aligned}$$

where we use the fact that $z_i \in [0, 1]$ and consequently $\mathbb{E}[|z_i|] \leq 1$.

Next, we define the sequence of random variables:

$$\begin{aligned}U_0 &= \mathbb{E}[f(\mathbf{z})], \\ U_1 &= \mathbb{E}[f(\mathbf{z})|z_1], \\ U_2 &= \mathbb{E}[f(\mathbf{z})|z_{1:2}], \\ &\vdots \\ U_i &= \mathbb{E}[f(\mathbf{z})|z_{1:i}], \\ &\vdots \\ U_n &= \mathbb{E}[f(\mathbf{z})|z_{1:n}],\end{aligned}$$

where $z_{i:j} := \{z_i, z_{i+1}, \dots, z_j\}$. Since $\mathbb{E}[|f(\mathbf{z})|] < \infty$, the sequence $\{U_0, U_1, \dots, U_n\}$ forms a Doob martingale.

The crux of the proof lies in using the Azuma-Hoeffding inequality:

Result (Azuma-Hoeffding). *If $\{X_k : k = 0, 1, 2, \dots\}$ is a martingale and there exist constants c_k such that $|X_k - X_{k-1}| \leq c_k$ for all k , then*

$$\mathbb{P}(|X_N - X_0| \geq t) \leq 2 \exp\left(\frac{-t^2}{2 \sum_{k=1}^N c_k^2}\right).$$

To use this inequality, we will show that there exist constants $c_1, c_2, \dots, c_n$ such that $|U_i - U_{i-1}| \leq c_i$ and

$$\sum_{i=1}^n c_i^2 = \frac{1}{2kn}$$

for some constant k .

Applying the Azuma-Hoeffding inequality will then yield

$$\begin{aligned} \mathbb{P}(|U_n - U_0| \geq t) &\leq 2 \exp\left(\frac{-t^2}{2(1/2kn)}\right) \\ \implies \mathbb{P}(|f(\mathbf{z})| \geq t) &\leq 2 \exp(-knt^2), \end{aligned}$$

where we used the facts that $U_n = f(\mathbf{z})$ and $U_0 = \mathbb{E}[f(\mathbf{z})] = 0$. This would complete the proof.

Part 1: Let us first consider $U_1 - U_0 = \mathbb{E}[f(\mathbf{z})|z_1]$ since $\mathbb{E}(f(\mathbf{z})) = 0$. Now,

$$\begin{aligned} \mathbb{E}[f(\mathbf{z})|z_1] &= \frac{1}{n} \mathbb{E}\left[\sum_{i=1}^n z_i a_i - b_i \middle| z_1\right] = \frac{1}{n} \left(\sum_{i=1}^n a_i \mathbb{E}[z_i|z_1] - b_i\right) \\ &= \frac{1}{n} \left(a_1 z_1 + \sum_{i=2}^n a_i \mathbb{P}(z_i = 1|z_1) - \sum_{i=1}^n b_i\right) \end{aligned}$$

Note that since $\mathbb{E}[f(\mathbf{z})] = 0$, hence, $\mathbb{E}[\sum_{i=1}^n a_i z_i] = \sum_{i=1}^n b_i$, using this, the above equality extends as

$$\begin{aligned} n(\mathbb{E}[f(\mathbf{z})|z_1]) &= a_1 z_1 + \sum_{i=2}^n a_i \mathbb{P}(z_i = 1|z_1) - \sum_{i=1}^n \mathbb{E}[a_i z_i] \\ &= a_1(z_1 - \mathbb{P}(z_1 = 1)) + \sum_{i=2}^n a_i(\mathbb{P}(z_i = 1|z_1) - \mathbb{P}(z_i = 1)). \end{aligned}$$

Our main goal is to bound the conditional expectation defined above. For $i \geq 2$, we bound the above probability difference:

$$\begin{aligned} |a_i(\mathbb{P}(z_i = 1|z_1) - \mathbb{P}(z_i = 1))| &\leq \max\{|a_i(\mathbb{P}(z_i = 1|z_1 = 1) - \mathbb{P}(z_i = 1))|, \\ &\quad |a_i(\mathbb{P}(z_i = 1|z_1 = 0) - \mathbb{P}(z_i = 1))|\}. \end{aligned}$$

Expressing in terms of y_i , the above is same as,

$$\max\{|a_i(\mathbb{P}(y_i \geq 0|y_1 \geq 0) - \mathbb{P}(y_i \geq 0))|, |a_i(\mathbb{P}(y_i \geq 0|y_1 \leq 0) - \mathbb{P}(y_i \geq 0))|\}.$$

Summing over all i , we get,

$$\begin{aligned} n(|\mathbb{E}[f(\mathbf{z})|z_1]|) &\leq |a_1| + \sum_{i=1}^n |a_i(\mathbb{P}(y_i \geq 0|y_1 \geq 0) - \mathbb{P}(y_i \geq 0))| \\ &\quad + \sum_{i=1}^n |a_i(\mathbb{P}(y_i \geq 0|y_1 \leq 0) - \mathbb{P}(y_i \geq 0))|, \end{aligned}$$

since, $|z_1 - \mathbb{P}(z_1 = 1)| \leq 1$.

Let $v_1^2, v_2^2, \dots, v_n^2$ be the diagonal elements of $\mathbf{A}_0^{-2}$, representing the variances of $y_1, y_2, \dots, y_n$. Define the normalized variables $\tilde{y}_i = y_i/v_i$, and let $\tilde{\mathbf{y}} = (\tilde{y}_1, \tilde{y}_2, \dots, \tilde{y}_n)^T$. Then,

$$\tilde{\mathbf{y}} \sim \mathcal{N}(\tilde{\boldsymbol{\mu}}, \boldsymbol{\Sigma}),$$

where $\boldsymbol{\mu} = \mathbf{A}_0^{-1} \mathbf{X} \boldsymbol{\beta}$ and $\tilde{\mu}_i = \mu_i/v_i$. Adel: I think it should be $\boldsymbol{\mu} = \mathbf{A}_0^{-1} \mathbf{X} \boldsymbol{\beta}$ and $\tilde{\mu}_i = \mu_i/v_i$. Correct. Changed it. The covariance matrix $\boldsymbol{\Sigma}$ is the correlation matrix corresponding to $\mathbf{A}_0^{-2}$. Adel: but you normalized by v_i 's Yes, v_i^2 's are the variances and v_i the standard deviation. Thus, $\text{Cov}(\tilde{y}_i, \tilde{y}_j) = \text{Cor}(y_i, y_j)$. Using Lemma 6, diagonal elements of $\mathbf{A}_0^{-2}$ are greater than 1, thus $v_i \geq 1$ for all i . Moreover, since

$$\Sigma_{ij} = \frac{(\mathbf{A}_0^{-2})_{ij}}{v_i v_j}, \quad \forall i \neq j,$$

we conclude that $\boldsymbol{\Sigma}$ is elementwise smaller than $\mathbf{A}_0^{-2}$ and consequently, $\|\boldsymbol{\Sigma}\|_\infty \leq \|\mathbf{A}_0^{-2}\|_\infty \leq 1/\epsilon^2$.

Since $\mathbf{A}_0^{-2} = \mathbf{I} + 2\rho \mathbf{W} + 3\rho^2 \mathbf{W}^2 + \dots$, it follows that $v_i \geq 1$ for all i . Adel: why?
Since $\mathbf{A}_0 = (\mathbf{I} - \rho \mathbf{W})$

Consider the summand $S_i = \mathbb{P}(y_i \geq 0 \mid y_1 \geq 0) - \mathbb{P}(y_i \geq 0)$. By the definition of $\tilde{\mathbf{y}}$, this simplifies to

$$S_i = \mathbb{P}(\tilde{y}_i \geq 0 \mid \tilde{y}_1 \geq 0) - \mathbb{P}(\tilde{y}_i \geq 0).$$

Define $\rho_i = \Sigma_{1i}$, the correlation between $\tilde{y}_i$ and $\tilde{y}_1$. Next, introduce the function g_i as

$$g_i(\rho) = \frac{\Phi_2(\mu_i, \mu_1, \rho)}{\Phi(\mu_1)},$$

where $\Phi_2(\mu_1, \mu_2, \rho)$ is defined as the distribution function of bivariate normal distribution with mean $\mathbf{0}$, variance $\mathbf{1}$ and correlation ρ ,

$$\Phi_2(\mu_1, \mu_2, \rho) = \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\mu_1} \int_{-\infty}^{\mu_2} \exp\left(-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right) dy dx.$$

Adel: Define Φ_2 Note the two observations below:

- Since $\tilde{y}_i$ and $\tilde{y}_1$ have correlation ρ_i , it follows that

$$g_i(\rho_i) = \frac{\mathbb{P}(\tilde{y}_i \geq 0, \tilde{y}_1 \geq 0)}{\mathbb{P}(\tilde{y}_1 \geq 0)} = \mathbb{P}(\tilde{y}_i \geq 0 \mid \tilde{y}_1 \geq 0).$$

- for $\rho = 0$, we get

$$g_i(0) = \frac{\Phi_2(\mu_i, \mu_1, 0)}{\Phi(\mu_1)} = \Phi(\mu_i) = \mathbb{P}(\tilde{y}_i \geq 0).$$

Thus

$$\begin{aligned}
|S_i| &= |\mathbb{P}(\tilde{y}_i \geq 0 | \tilde{y}_1 \geq 0) - \mathbb{P}(\tilde{y}_i \geq 0)| = |g(\rho_i) - g(0)| \\
&\leq \int_0^{\rho_i} |g'_i(t)| dt = \frac{1}{\Phi(\mu_1)} \int_0^{\rho_i} \phi_2(\mu_i, \mu_1, t) dt \\
&= \frac{1}{\Phi(\mu_1)} \int_0^{\rho_i} \frac{1}{2\pi\sqrt{1-t^2}} \exp\{-(\mu_i^2 - 2t\mu_1\mu_i + \mu_1^2)/2(1-t^2)\} dt \\
&\leq \frac{1}{\Phi(\mu_1)} \int_0^{\rho_i} \frac{1}{2\pi\sqrt{1-t^2}} dt = \frac{\arcsin(\rho_i)}{\Phi(\mu_1)2\pi},
\end{aligned}$$

Adel: third line above, ρ should be t ? correct, changed where the equality in the third Adel: third? changed line is a standard result (Sungur, 1990; Sibuya et al., 1960). Hence,

$$\sum_{i=1}^n |a_i(\mathbb{P}(y_i \geq 0 | y_1 \geq 0) - \mathbb{P}(y_i \geq 0))| = \sum_{i=1}^n a_i |S_i| \leq \sum_{i=1}^n \frac{a_i \arcsin(\rho_i)}{2\pi\Phi(\mu_1)}$$

Using $a_i \leq a$ and $|\tilde{\mu}_1| \leq C_x C_\beta s_0 / \epsilon$ (proved in Lemma 4), we have

$$|\mu_1| = \left| \frac{\tilde{\mu}_1}{v_1} \right| \leq |\tilde{\mu}_1| \leq \frac{C_x C_\beta s_0}{\epsilon}.$$

Thus,

$$\Phi(\mu_1) \geq \Phi\left(-\frac{C_x C_\beta s_0}{\epsilon}\right) := C_\Phi.$$

With this, we obtain

$$\sum_{i=1}^n |a_i(\mathbb{P}(y_i \geq 0 | y_1 \geq 0) - \mathbb{P}(y_i \geq 0))| \leq \frac{a}{2\pi C_\Phi} \sum_{i=1}^n \arcsin(\rho_i).$$

Since $\rho_i = \Sigma_{1i} \leq (\mathbf{A}_0^{-2})_{1i}$ and $\|\mathbf{A}_0^{-2}\|_\infty \leq 1/\epsilon^2$ (Lemma 6), it follows that $\sum_{i=1}^n \rho_i \leq \|\Sigma\|_\infty \leq 1/\epsilon^2$ Adel: Why? Since Σ is elementwise smaller than $\mathbf{A}_0^{-2}$ and $\|\mathbf{A}_0^{-2}\|_\infty \leq 1/\epsilon^2$. Given that $\arcsin(t)/t$ is maximized at $t = 1$, we get $\sum_{i=1}^n \arcsin(\rho_i) \leq \pi/2\epsilon^2$.

Substituting this into the previous bound, we finally arrive at

$$\sum_{i=1}^n |a_i(\mathbb{P}(y_i \geq 0 | y_1 \geq 0) - \mathbb{P}(y_i \geq 0))| \leq \frac{a}{4C_\Phi \epsilon^2}.$$

Now, assuming $g(\rho) = \Phi_2(\mu_i, -\mu_1, \rho)/\Phi(-\mu_1)$, we obtain the analogous inequality

$$\sum_{i=1}^n |a_i(\mathbb{P}(y_i \geq 0 | y_1 \leq 0) - \mathbb{P}(y_i \geq 0))| \leq \frac{a}{4C_\Phi \epsilon^2}.$$

Combining both, we conclude that

$$n(|\mathbb{E}[f(\mathbf{z}) | \mathbf{z}_1]|) \leq |a_1| + 2\frac{a}{4C_\Phi \epsilon^2} \leq a \left(1 + \frac{1}{2C_\Phi \epsilon^2}\right).$$

By setting $c_1 = \frac{a}{n} \left(1 + \frac{1}{2C_\Phi \epsilon^2}\right)$, we have $U_1 - U_0 = \mathbb{E}[f(\mathbf{z}) \mid z_1] \leq c_1$.

Adel: Conclude Part 1, by saying $c_1 = \dots$

Part j : Next at step j , we need to bound $|U_{j+1} - U_j| = |\mathbb{E}[f(\mathbf{z}) \mid z_{1:j+1}] - \mathbb{E}[f(\mathbf{z}) \mid z_{1:j}]|$.

$$n(\mathbb{E}[f(\mathbf{z}) \mid z_{1:j+1}]) = \sum_{i=1}^{j+1} a_i z_i + \sum_{i=j+2}^n a_i \mathbb{P}(z_i = 1 \mid z_{1:j+1}) - \sum_{i=1}^n b_i,$$

and similarly,

$$n(\mathbb{E}[f(\mathbf{z}) \mid z_{1:j}]) = \sum_{i=1}^j a_i z_i + \sum_{i=j+1}^n a_i \mathbb{P}(z_i = 1 \mid z_{1:j}) - \sum_{i=1}^n b_i.$$

Thus,

$$\begin{aligned} n(\mathbb{E}[f(\mathbf{z}) \mid z_{1:j+1}] - \mathbb{E}[f(\mathbf{z}) \mid z_{1:j}]) &= a_{j+1}(z_{j+1} - \mathbb{P}(z_{j+1} = 1 \mid z_{1:j})) \\ &\quad + \sum_{i=j+2}^n a_i (\mathbb{P}(z_i = 1 \mid z_{1:j+1}) - \mathbb{P}(z_i = 1 \mid z_{1:j})) \\ &\leq a_{j+1} + \sum_{i=j+2}^n a_i (\mathbb{P}(z_i = 1 \mid z_{1:j+1}) - \mathbb{P}(z_i = 1 \mid z_{1:j})) \\ &\leq a_{j+1} + \sum_{i=j+2}^n a_i (\mathbb{P}(z_i = 1 \mid z_{j+1} = 1, z_{1:j}) - \mathbb{P}(z_i = 1 \mid z_{1:j})) \\ &\quad + \sum_{i=j+2}^n a_i (\mathbb{P}(z_i = 1 \mid z_{j+1} = 0, z_{1:j}) - \mathbb{P}(z_i = 1 \mid z_{1:j})). \end{aligned}$$

We here want to exploit the fact that the conditional distribution of $y_{j+1:n} \mid y_{1:j}$ is jointly normal.

$$\mathbf{y}_{j+1:n} \mid \mathbf{y}_{1:j} \sim \mathcal{N} \left(\underbrace{\boldsymbol{\mu}_2 + \mathbf{A}_{21} \mathbf{A}_{11}^{-1} (\mathbf{y}_{1:j} - \boldsymbol{\mu}_1)}_{\tilde{\boldsymbol{\mu}}}, \underbrace{\mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}}_{\tilde{\mathbf{A}}} \right),$$

where $\boldsymbol{\mu}_1, \boldsymbol{\mu}_2$ and $\mathbf{A}_{11}, \mathbf{A}_{12}, \mathbf{A}_{21}, \mathbf{A}_{22}$ are defined as

$$\boldsymbol{\mu}_1 = (\mathbf{A}_0^{-1} \mathbf{X} \boldsymbol{\beta}_0)_{1:j}, \quad \boldsymbol{\mu}_2 = (\mathbf{A}_0^{-1} \mathbf{X} \boldsymbol{\beta}_0)_{j+1:n}, \quad \mathbf{A}_0^{-2} = \left[\begin{array}{c|c} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \hline \mathbf{A}_{21} & \mathbf{A}_{22} \end{array} \right],$$

$\mathbf{A}_{11} \in \mathbb{R}^{j \times j}$, $\mathbf{A}_{12}, \mathbf{A}_{21} \in \mathbb{R}^{j \times (n-j)}$, $\mathbf{A}_{22} \in \mathbb{R}^{(n-j) \times (n-j)}$. The conditional covariance matrix, $\tilde{\mathbf{A}} = \mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$, follows two important properties:

1. **Diagonal elements of $\tilde{\mathbf{A}}$ are bounded below**

Note that for any l , $(\tilde{\mathbf{A}})_l = \text{Var}(y_l|y_{1:j})$. The proof requires two important facts, first $\text{Var}(y_l|y_{1:j}) \geq \text{Var}(y_l|y_{-l})$, since the second random variable is conditioned on more variables. Second, $\text{Var}(\mathbf{y}) = \mathbf{A}_0^{-2}$, hence $\text{Var}(y_l|y_{-l}) = 1/(\mathbf{A}_0^2)_l$ since diagonal of the precision matrix is same as the conditional variance.

Adel: needs further explanation Since $\|\mathbf{A}_0\|_\infty = \|\mathbf{I} - \rho\mathbf{W}\|_\infty \leq 1 + \rho \leq 2 - \epsilon$, L_1 and consequently L_2 norm of each row is bounded by $(2 - \epsilon)^2$ **Adel: why squared?**. This implies, $(\mathbf{A}_0^2)_l \leq (1 - \epsilon)^2$ **Adel: why?** and

$$(\tilde{\mathbf{A}})_l = \text{Var}(y_l|y_{1:j}) \geq \text{Var}(y_l|y_{-l}) \geq \frac{1}{(2 - \epsilon)^2}.$$

Note that for any l , $(\tilde{\mathbf{A}})_l = \text{Var}(y_l|y_{1:j})$. The proof requires two important facts, first $\text{Var}(y_l|y_{1:j}) \geq \text{Var}(y_l|y_{-l})$, since the second random variable is conditioned on more variables.

Second, since $\text{Var}(\mathbf{y}) = \mathbf{A}_0^{-2}$, the precision matrix is $\mathbf{A}_0^2$. In a multivariate normal distribution the conditional variance of a single variable given all the others is the reciprocal of the corresponding diagonal element of the precision matrix, thus we get $\text{Var}(y_l|y_{-l}) = 1/(\mathbf{A}_0^2)_l$.

Finally, $\|\mathbf{A}_0\|_\infty = \|\mathbf{I} - \rho\mathbf{W}\|_\infty \leq 1 + |\rho| \leq 2 - \epsilon$, thus $\mathbf{a}_l^\top$, the l -th row of $\mathbf{A}_0$ has $\|\mathbf{a}_l\|_1 \leq 2 - \epsilon$ and consequently $\|\mathbf{a}_l\|_2 \leq \|\mathbf{a}_l\|_1 \leq 2 - \epsilon$. This implies, $(\mathbf{A}_0^2)_l = \mathbf{a}_l^\top \mathbf{a}_l = \|\mathbf{a}_l\|_2^2 \leq (2 - \epsilon)^2$. and

$$(\tilde{\mathbf{A}})_l = \text{Var}(y_l|y_{1:j}) \geq \text{Var}(y_l|y_{-l}) = \frac{1}{(\mathbf{A}_0^2)_l} \geq \frac{1}{(2 - \epsilon)^2}.$$

2. Covariance matrix $\tilde{\mathbf{A}}$ has bounded row sum: $\|\tilde{\mathbf{A}}\|_\infty \leq C_A$

We first see that $\|\tilde{\mathbf{A}}\|_\infty \leq \|\mathbf{A}_{22}\|_\infty + \|\mathbf{A}_{21}\|_\infty \|\mathbf{A}_{11}^{-1}\|_\infty \|\mathbf{A}_{12}\|_\infty$. Since $\mathbf{A}_{22}$, $\mathbf{A}_{21}$ and $\mathbf{A}_{12}$ are blocks **Adel: What are these blocks?** of $\mathbf{A}_0^{-2}$ and $\|\mathbf{A}_0^{-2}\|_\infty \leq 1/\epsilon^2$, thus $\|\tilde{\mathbf{A}}\|_\infty \leq 1/\epsilon^2 + \|\mathbf{A}_{11}^{-1}\|_\infty/\epsilon^4$. Consider arbitrary $\mathbf{m} \in \mathbb{R}^j$ and $\tilde{\mathbf{m}} = (\mathbf{m}^T, 0^T)^T$. Since $\|\mathbf{A}_0^2\|_\infty \leq 2 - \epsilon$, we have

$$\begin{aligned} \frac{\|\mathbf{A}_0^2(\mathbf{A}_0^{-2}\tilde{\mathbf{m}})\|_\infty}{\|\mathbf{A}_0^{-2}\tilde{\mathbf{m}}\|_\infty} &\leq \|\mathbf{A}_0^2\|_\infty \leq 2 - \epsilon \\ \implies \|\tilde{\mathbf{m}}\|_\infty &\leq (2 - \epsilon)\|\mathbf{A}_0^{-2}\tilde{\mathbf{m}}\|_\infty \end{aligned}$$

Since $\tilde{\mathbf{m}} = (\mathbf{m}^T, 0^T)^T$, implies $\|\tilde{\mathbf{m}}\|_\infty = \|\mathbf{m}\|_\infty$ and $\|\mathbf{A}_0^{-2}\tilde{\mathbf{m}}\|_\infty = \max\{\|\mathbf{A}_{11}\mathbf{m}\|_\infty, \|\mathbf{A}_{21}\mathbf{m}\|_\infty\} \geq \|\mathbf{A}_{11}\mathbf{m}\|_\infty$. Thus we have for all $\mathbf{m} \in \mathbb{R}^j$,

Adel: the following does not result from above!

$$\|\mathbf{m}\|_\infty \leq (2 - \epsilon)\|\mathbf{A}_{11}\mathbf{m}\|_\infty.$$

Now, for any $\mathbf{k} \in \mathbb{R}^j$, let $\mathbf{m} = \mathbf{A}_{11}^{-1}\mathbf{k}$, this means,

$$\|\mathbf{A}_{11}^{-1}\mathbf{k}\|_\infty = \|\mathbf{m}\|_\infty \leq (2 - \epsilon)\|\mathbf{A}_{11}\mathbf{m}\|_\infty = (2 - \epsilon)\|\mathbf{k}\|_\infty.$$

Since this is true for any arbitrary $\mathbf{k}$, hence $\|\mathbf{A}_{11}^{-1}\|_\infty \leq 2 - \epsilon$ which completes the proof.

First note that $\tilde{\mathbf{A}}$ is the Schur complement of $\mathbf{A}_{22}$ in $\mathbf{A}_0^{-2}$. We use the following result in [Horn and Zhang \(2005\)](#) Theorem 1.2 .

Result. *Let $\mathbf{M}$ be a matrix partitioned as*

$$\mathbf{M} = \left[\begin{array}{c|c} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \hline \mathbf{M}_{21} & \mathbf{M}_{22} \end{array} \right],$$

and suppose both $\mathbf{M}$ and $\mathbf{M}_{22}$ are nonsingular. Then the Schur complement of $\mathbf{M}_{22}$ in $\mathbf{M}$ denoted as $\mathbf{M}/\mathbf{M}_{22}$ is nonsingular and

$$(\mathbf{M}^{-1})_{11} = (\mathbf{M}/\mathbf{M}_{22})^{-1}, \text{ where } \mathbf{M}^{-1} = \left[\begin{array}{c|c} (\mathbf{M}^{-1})_{11} & (\mathbf{M}^{-1})_{12} \\ \hline (\mathbf{M}^{-1})_{21} & (\mathbf{M}^{-1})_{22} \end{array} \right].$$

Since $\mathbf{A}_0^{-2}$ is a covariance matrix, both $\mathbf{A}_{22}$ and $\mathbf{A}_0^{-2}$ are nonsingular, so we can apply the result directly. The result thus implies,

$$\tilde{\mathbf{A}} = \mathbf{A}_0^{-2}/\mathbf{A}_{22} = \{(\mathbf{A}_0^2)_{11}\}^{-1}, \text{ where } \mathbf{A}_0^2 = \left[\begin{array}{c|c} (\mathbf{A}_0^2)_{11} & (\mathbf{A}_0^2)_{12} \\ \hline (\mathbf{A}_0^2)_{21} & (\mathbf{A}_0^2)_{22} \end{array} \right].$$

Let $\mathbf{B} = (\mathbf{A}_0)_{11}$, then $(\mathbf{A}_0^2)_{11} = \mathbf{B}^2 + \rho_0^2 \mathbf{W}_{12} \mathbf{W}_{21}$ since $\mathbf{A}_0 = \mathbf{I} - \rho_0 \mathbf{W}$. Also, since $\mathbf{W}$ is symmetric, $(\mathbf{A}_0^2)_{11} = \mathbf{B}^2 + \rho_0^2 \mathbf{W}_{21}^\top \mathbf{W}_{21} = \mathbf{B}(\mathbf{I} + \mathbf{K})\mathbf{B}$ where $\mathbf{K} = \mathbf{B}^{-1} \rho_0^2 \mathbf{W}_{21}^\top \mathbf{W}_{21} \mathbf{B}^{-1} = (\rho_0 \mathbf{W}_{21} \mathbf{B}^{-1})^\top (\rho_0 \mathbf{W}_{21} \mathbf{B}^{-1})$, thus $\mathbf{K}$ is positive semi-definite and the eigenvalues of $(\mathbf{I} + \mathbf{K})$ are all greater than 1.

Similar to the first part, consider the function **Adel: you condition on $y_{1:j}$ or $\tilde{y}_{1:j}$?**

$$g_i(\rho) = \mathbb{P}(\tilde{y}_i \geq 0, \tilde{y}_{j+1} \geq 0 | y_{1:j}),$$

where $\tilde{y}_i | y_{1:j}$, $\tilde{y}_{j+1} | y_{1:j}$ have the same marginal distribution as $y_i | y_{1:j}$, $y_{j+1} | y_{1:j}$ respectively **Adel: I don't think it is true, only the sign of $\tilde{y}_i$ and y_i matches and that's what you need.**, and $\tilde{y}_i | y_{1:j}$, $\tilde{y}_{j+1} | y_{1:j}$ have correlation ρ . Alternatively, we can rewrite the function as

$$g_i(\rho) = \Phi(\mu_i, \mu_{j+1}, \rho),$$

where, $\mu_i = \tilde{\mu}_i / (\sqrt{(\tilde{\mathbf{A}})_{ii}})$ and $\mu_{j+1} = \tilde{\mu}_{j+1} / (\sqrt{(\tilde{\mathbf{A}})_{j+1,j+1}})$. This follows by normalizing $\tilde{y}_i, \tilde{y}_{j+1}$ by their standard deviations.

Note two things, **Adel: informal writing** first if $\rho_i = (\tilde{\mathbf{A}})_{1,i-j} / ((\tilde{\mathbf{A}})_{11}(\tilde{\mathbf{A}})_{i-j,i-j})$, then $g_i(\rho_i) = \mathbb{P}(y_i \geq 0, y_{j+1} \geq 0 | y_{1:j})$, since $(\tilde{y}_i, \tilde{y}_{j+1} \equiv y_i, y_{j+1})$ in that case.

Second, $g_i(0) = \mathbb{P}(\tilde{y}_i \geq 0, \tilde{y}_{j+1} \geq 0 | y_{1:j}) = \mathbb{P}(\tilde{y}_i \geq 0 | y_{1:j})\mathbb{P}(\tilde{y}_{j+1} \geq 0 | y_{1:j})$ since correlation 0 implies independence in joint normal situation. Since marginally, $\tilde{y}_i \equiv y_i$ and $\tilde{y}_{j+1} \equiv y_{j+1}$, thus $g_i(0) = \mathbb{P}(\tilde{y}_i \geq 0 | y_{1:j})\mathbb{P}(\tilde{y}_{j+1} \geq 0 | y_{1:j}) = \mathbb{P}(y_i \geq 0 | y_{1:j})\mathbb{P}(y_{j+1} \geq 0 | y_{1:j})$. Combining both the terms, we get

$$g_i(\rho_i) - g_i(0) = \mathbb{P}(y_i \geq 0, y_{j+1} \geq 0 | y_{1:j}) - \mathbb{P}(y_i \geq 0 | y_{1:j})\mathbb{P}(y_{j+1} \geq 0 | y_{1:j})$$

Using similar argument as in part 1,

$$\begin{aligned} |\mathbb{P}(y_i \geq 0, y_{j+1} \geq 0 | y_{1:j}) - \mathbb{P}(y_i \geq 0 | y_{1:j})\mathbb{P}(y_{j+1} \geq 0 | y_{1:j})| &= |g_i(\rho_i) - g_i(0)| \\ &\leq \int_0^{\rho_i} |g'_i(t)| dt = \int_0^{\rho_i} \phi_2(\mu_i, \mu_{j+1}, t) dt \\ &= \int_0^{\rho_i} \frac{1}{2\pi\sqrt{1-t^2}} \exp\{-(\mu_i^2 - 2\rho\mu_{j+1}\mu_i + \mu_{j+1}^2)/2(1-t^2)\} dt \\ &\leq \int_0^{\rho_i} \frac{1}{2\pi\sqrt{1-t^2}} dt = \frac{\arcsin(\rho_i)}{2\pi}, \end{aligned}$$

Since the inequality is true for all possible $y_{1:j}$, hence we get

$$|\mathbb{P}(y_i \geq 0, y_{j+1} \geq 0 | z_{1:j}) - \mathbb{P}(y_i \geq 0 | y_{1:j})\mathbb{P}(y_{j+1} \geq 0 | z_{1:j})| \leq \frac{\arcsin(\rho_i)}{2\pi}.$$

Dividing both sides by $\mathbb{P}(y_{j+1} \geq 0 | z_{1:j})$, this get simplified as

$$\begin{aligned} &\left| \frac{\mathbb{P}(y_i \geq 0, y_{j+1} \geq 0 | z_{1:j})}{\mathbb{P}(y_{j+1} \geq 0 | z_{1:j})} - \mathbb{P}(y_i \geq 0 | y_{1:j}) \right| \\ &= \frac{|\mathbb{P}(y_i \geq 0 | y_{j+1} \geq 0, z_{1:j}) - \mathbb{P}(y_i \geq 0 | y_{1:j})\mathbb{P}(y_{j+1} \geq 0 | z_{1:j})|}{\mathbb{P}(y_{j+1} \geq 0 | z_{1:j})} \\ &\leq \frac{\arcsin(\rho_i)}{2\pi\mathbb{P}(y_{j+1} \geq 0 | z_{1:j})} \end{aligned}$$

Adel: in the second line the last term $\mathbb{P}(y_{j+1} \geq 0 | z_{1:j})$ is extra **Corrected**. Since, diagonal element of $\tilde{\mathbf{A}}$ are lower bounded by c_A and $\|\tilde{\mathbf{A}}\|_\infty \leq C_A$, hence $\sum_{i=j+2}^n \rho_i \leq C_A/c_A$. Using this, we can bound

$$\sum_{i=j+2}^n |a_i(\mathbb{P}(z_i = 1 | z_{j+1} = 1, z_{1:j}) - \mathbb{P}(z_i = 1 | z_{1:j}))| \leq \frac{aC_A}{4c_A\mathbb{P}(y_{j+1} \geq 0 | z_{1:j})},$$

using the same trick as in part 1.

Thus,

$$|\mathbb{E}[f(\mathbf{z})|z_{1:j+1}] - \mathbb{E}[f(\mathbf{z})|z_{1:j}]| \leq a \left(1 + \frac{C_A}{2c_A \mathbb{P}(y_{j+1} \geq 0|z_{1:j})} \right)$$

Finally with the following lemma we show a lower bound C_z on $\mathbb{P}(y_{j+1} \geq 0|z_{1:j})$.

Lemma 7. *Let $\mathbf{y} \sim \mathcal{N}(\mathbf{A}_0^{-1} \mathbf{X} \boldsymbol{\beta}_0, \mathbf{A}_0^{-2})$, and $\mathbf{z} = \mathbf{1}(\mathbf{y} > 0)$, then there exists constant C_z , such that*

$$\mathbb{P}(y_{j+1} \geq 0|z_{1:j}) \geq C_z,$$

for all j and $z_{1:j}$.

This completes the proof as we upper bound each summand by a constant $a(1 + C_A/2c_A c_z)$, and $\sum_{i=1}^n c_i^2 = \mathcal{O}(n)$.

B.4 Proof of Lemma 4

Recall that the pseudo log-likelihood is defined as Adel: wasn't it called ℓ_n before, why is it changed to $\tilde{\ell}_n$? Right, changed.

$$\ell_n(\rho, \boldsymbol{\beta}) = -\frac{1}{n} \sum_{i=1}^n \left\{ z_i \log \Phi \left(\frac{\mathbf{e}_i^T \mathbf{A}^{-1} \mathbf{X} \boldsymbol{\beta}}{\|\mathbf{e}_i^T \mathbf{A}^{-1}\|} \right) + (1 - z_i) \log \left(1 - \Phi \left(\frac{\mathbf{e}_i^T \mathbf{A}^{-1} \mathbf{X} \boldsymbol{\beta}}{\|\mathbf{e}_i^T \mathbf{A}^{-1}\|} \right) \right) \right\},$$

The partial derivative is thus given by:

$$\begin{aligned} \frac{\partial \ell_n}{\partial \beta_j} \Big|_{\gamma_0} &= -\frac{1}{n} \sum_{i=1}^n z_i \frac{\phi(t_i)}{\Phi(t_i)} t_{ij} + z_i \frac{\phi(t_i)}{1 - \Phi(t_i)} t_{ij} - \frac{\phi(t_i)}{1 - \Phi(t_i)} t_{ij} \\ &= -\frac{1}{n} \sum_{i=1}^n z_i \frac{\phi(t_i) t_{ij}}{\Phi(t_i)(1 - \Phi(t_i))} - \frac{\phi(t_i)}{1 - \Phi(t_i)} t_{ij}, \end{aligned}$$

where t_{ij} and t_i are defined as, Adel: What is $\mathbf{A}_0$? same as $\mathbf{A}$? Is $\mathbf{X}_j$ the j -th column of $\mathbf{X}$? it is not defined Defined $\mathbf{A}_0$ and $\mathbf{X}_j$.

$$\begin{aligned} t_{ij} &= \frac{\partial}{\partial \beta_j} \left(\frac{\mathbf{e}_i^T \mathbf{A}^{-1} \mathbf{X} \boldsymbol{\beta}}{\|\mathbf{e}_i^T \mathbf{A}^{-1}\|} \right) \Big|_{\gamma_0} = \frac{\mathbf{e}_i^T \mathbf{A}_0^{-1} \mathbf{X}_j}{\|\mathbf{e}_i^T \mathbf{A}_0^{-1}\|}, \quad \forall 1 \leq i \leq n, 1 \leq j \leq p \\ t_i &= \sum_{j=1}^p t_{ij} \beta_{0j} = \frac{\mathbf{e}_i^T \mathbf{A}_0^{-1} \mathbf{X} \boldsymbol{\beta}_0}{\|\mathbf{e}_i^T \mathbf{A}_0^{-1}\|}, \quad \forall i \leq n, \end{aligned}$$

where $\mathbf{A}_0 = \mathbf{I} - \rho_0 \mathbf{W}$ and $\mathbf{X}_j$ the j -th column of $\mathbf{X}$. Next, we establish that t_{ij} is bounded for all i and j .

First, using binomial expansion for $(\mathbf{I} - \rho_0 \mathbf{W})^{-1}$, we can write:

$$\mathbf{A}_0^{-1} = (\mathbf{I} - \rho_0 \mathbf{W})^{-1} = \mathbf{I} + \rho_0 \mathbf{W} + \rho_0^2 \mathbf{W}^2 + \dots$$

Since induced matrix norms are sub-multiplicative Adel: we never defined these norms Added a section to define these norms., hence $\|\mathbf{W}\|_\infty \leq 1$, implies $\|\mathbf{W}^i\|_\infty \leq 1$ for all $i \geq 1$. Thus,

$$\|\mathbf{A}_0^{-1}\|_\infty \leq 1 + \rho_0 + \rho_0^2 + \dots = 1/(1 - \rho_0) \leq 1/\epsilon.$$

Using Lemma 6, we have $\|\mathbf{A}_0^{-1}\|_\infty \leq (1 - \rho_0)^{-1} \leq 1/\epsilon$. Since induced matrix norms are sub-multiplicative, $\|\mathbf{A}_0^{-1} \mathbf{X}_j\|_\infty \leq \|\mathbf{A}_0^{-1}\|_\infty \|\mathbf{X}_j\|_\infty \leq C_x/\epsilon$ implying that $|\mathbf{e}_i^\top \mathbf{A}_0^{-1} \mathbf{X}_j| \leq C_x/\epsilon$. Also note that $\|\mathbf{e}_i^\top \mathbf{A}_0^{-1}\| \geq |(\mathbf{A}_0^{-1})_{ii}| \geq 1$

Adel: You have used both C_X and C_x make it consistent. Changed everything to C_x .

Consequently, since $t_i = \sum_{j=1}^p t_{ij} \beta_{0j}$, we have: Adel: What about the denominator $\|\mathbf{e}_i^\top \mathbf{A}_0^{-1}\|$? $\|\mathbf{e}_i^\top \mathbf{A}_0^{-1}\|$

$$t_i \leq \sum_{j=1}^{s_0} \frac{C_x}{\epsilon} C_\beta = \frac{C_x C_\beta s_0}{\epsilon},$$

using Assumption 3 ($|\beta_{0j}| \leq C_\beta$ and β_{0j} is nonzero for at most s_0 elements).

Similar to $\partial \tilde{\ell}_n / \partial \beta_j|_{\gamma_0}$, $\partial \tilde{\ell}_n / \partial \rho|_{\gamma_0}$ can be simplified as

$$\begin{aligned} \frac{\partial \tilde{\ell}_n}{\partial \rho} \Big|_{\gamma_0} &= -\frac{1}{n} \sum_{i=1}^n z_i \frac{\phi(t_i)}{\Phi(t_i)} t_{i0} + z_i \frac{\phi(t_i)}{1 - \Phi(t_i)} t_{i0} - \frac{\phi(t_i)}{1 - \Phi(t_i)} t_{i0} \\ &= -\frac{1}{n} \sum_{i=1}^n z_i \frac{\phi(t_i) t_{i0}}{\Phi(t_i)(1 - \Phi(t_i))} - \frac{\phi(t_i)}{1 - \Phi(t_i)} t_{i0}, \end{aligned}$$

where, t_{i0} is defined as, Adel: Explain the inequality below. Also W should be $\mathbf{W}$ throughout

$$t_{i0} = \frac{\partial}{\partial \rho} \left(\frac{\mathbf{e}_i^\top \mathbf{A}^{-1} \mathbf{X} \beta}{\|\mathbf{e}_i^\top \mathbf{A}^{-1}\|} \right) \Big|_{\gamma_0} \leq \frac{\|\mathbf{e}_i^\top \mathbf{A}_0^{-1}\| \mathbf{e}_i^\top \mathbf{A}_0^{-2} \mathbf{W} \mathbf{X} \beta_0}{\|\mathbf{e}_i^\top \mathbf{A}_0^{-1}\|^2} = \frac{\mathbf{e}_i^\top \mathbf{A}_0^{-2} \mathbf{W} \mathbf{X} \beta_0}{\|\mathbf{e}_i^\top \mathbf{A}_0^{-1}\|}$$

Similar to earlier, since

$$\mathbf{A}_0^{-2} = \mathbf{I} + 2\rho \mathbf{W} + 3\rho^2 \mathbf{W}^2 + \dots$$

hence $\|\mathbf{A}_0^{-2}\|_\infty \leq \sum_{i \geq 0} (i+1) \rho^i = 1/(1 - \rho)^2 \leq 1/\epsilon^2$. We also use the fact that $\|\mathbf{W}\|_\infty \leq 1$ and $\|\mathbf{X} \beta_0\|_\infty \leq C_x C_\beta s_0$. Thus again using the sub-multiplicative property, $\|\mathbf{A}_0^{-2} \mathbf{W} \mathbf{X} \beta_0\|_\infty \leq C_x C_\beta s_0$ and thus,

$$\frac{\mathbf{e}_i^\top \mathbf{A}_0^{-2} \mathbf{W} \mathbf{X} \beta_0}{\|\mathbf{e}_i^\top \mathbf{A}_0^{-1}\|} \leq \frac{C_x C_\beta s_0}{\epsilon^2}.$$

We can rewrite all the partial derivatives as

$$\frac{\partial \tilde{\ell}_n}{\partial \beta_j} \Big|_{\gamma_0} = -\frac{1}{n} \sum_{i=1}^n (z_i a_{ij} - b_{ij}) \quad \frac{\partial \tilde{\ell}_n}{\partial \rho} \Big|_{\gamma_0} = -\frac{1}{n} \sum_{i=1}^n (z_i a_{i0} - b_{i0}),$$

where

$$a_{ij} = \frac{\phi(t_i)t_{ij}}{\Phi(t_i)(1 - \Phi(t_i))}, \quad b_{ij} = \frac{\phi(t_i)}{1 - \Phi(t_i)}t_{ij}.$$

Note that since t_{ij} 's and t_i 's are bounded, a_{ij} and b_{ij} are bounded consequently as well
Adel: What bounded means here? I assume s_0 can grow as a function of p and n . Are we assuming p fixed and $n \rightarrow \infty$? We have not stated it anywhere.. We assume that $p = o(n)$ and s_0 is fixed. Or we can assume that $\|\beta\|_1$ is bounded and then $C_\beta s_0$ is bounded.

B.5 Proof of Lemma 7

We first note a few.

- The mean of $\mathbf{y}$, $\boldsymbol{\mu} := \mathbf{A}_0^{-1} \mathbf{X} \beta_0$, is bounded with respect to infinity norm, that is $\|\boldsymbol{\mu}\|_\infty \leq C_x C_\beta / \epsilon$ for some constant C_μ
- The covariance of $\mathbf{Y}$, $\mathbf{A}_0^{-2}$ has bounded row sum, $\|\mathbf{A}_0^{-2}\|_\infty \leq 1/\epsilon^2$.
- The pairwise correlation of y_i, y_j for any $i \neq j$ can be upper bounded by $2(1 - \epsilon)/(2 - 2\epsilon + \epsilon^2)$.

The first follows from the fact that $\|\mathbf{A}_0^{-1}\|_\infty \leq 1/\epsilon$, $\|\mathbf{X}\|_\infty \leq C_x$ and $\|\beta\|_\infty \leq C_\beta$. Similarly, the second follows from $\|\mathbf{A}_0^{-1}\|_\infty \leq 1/\epsilon$. Finally, pairwise correlation is highest for y_i and y_j when $W_{ij} = 1$. Thus all the other elements in the i th and j th column and row is 0. Thus, by rearranging $\mathbf{W}$, we can convert $\mathbf{W}$ into block diagonal matrices. **Adel:** I don't get this The block corresponding to i, j in $\mathbf{A}_0^{-2}$ is

$$(\mathbf{A}_0^{-2})_{ij} = \frac{1}{(1 - \rho_0)^2} \begin{pmatrix} 1 + \rho_0^2 & 2\rho_0 \\ 2\rho_0 & 1 + \rho_0^2 \end{pmatrix}.$$

Adel: What is ρ_0 ? This gives us the bound on correlation, as $2\rho_0/(1 + \rho_0^2)$ is maximized at $2(1 - \epsilon)/(2 - 2\epsilon + \epsilon^2)$. **Adel:** Why?

Since $\|\boldsymbol{\mu}\|_\infty \leq C_x C_\beta / \epsilon$, without loss of generality, assume that $\mathbf{y}$ has mean $\mathbf{0}$. **Adel:** Why? Also note that the probability we are trying to bound is independent of the scale of $\mathbf{y}$, hence let us assume that $\mathbf{y} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma}$ is the correlation matrix corresponding to $\mathbf{A}_0^{-2}$. **Adel:** You mean $\boldsymbol{\Sigma} = \mathbf{A}_0^{-2}$? Since the pairwise correlation is bounded, so are the elements of $\boldsymbol{\Sigma}$.

Also, since $\|\mathbf{A}_0^{-2}\|_\infty \leq 1/\epsilon^2$ and the diagonal elements are greater than 1, hence $\|\boldsymbol{\Sigma}\|_\infty \leq 1/\epsilon^2$. Also, to simplify, let us assume that $\rho \geq 0$, thus all the elements of $\boldsymbol{\Sigma}$ are positive.

Under the assumption that $\rho \geq 0$, then the $\mathbb{P}(y_{j+1} \geq 0 | z_{1:j})$ is minimized when $z_{1:j} = \mathbf{0}$ or $y_i \leq 0$ for all $i \leq j$. Let $\rho_i = \Sigma_{j+1,i}$ for $i \leq j$, then we can restructure the worst case problem as **Adel:** What is U ? is it y_{j+1} ?

$$y_i = \sqrt{\rho_i}U + \sqrt{1 - \rho_i}V_i \quad \text{where } U, V_i \sim \mathcal{N}(0, 1) \quad \text{Adel : independent?},$$

and this conserves the correlation structure. Also as described earlier, $\rho_i \leq 2(1 - \epsilon)/(2 - 2\epsilon + \epsilon^2)$ and $\sum_{i=1}^j \rho_i \leq 1/\epsilon^2$.

We are interested in $\mathbb{P}(y_{j+1} \geq 0 | y_i < 0 \text{ for all } i \leq j)$,

$$\mathbb{P}(y_{j+1} \geq 0 | y_1 < 0, y_2 < 0, \dots, y_j < 0) = \frac{\mathbb{P}(y_{j+1} \geq 0, y_1 < 0, y_2 < 0, \dots, y_j < 0)}{\mathbb{P}(y_1 < 0, y_2 < 0, \dots, y_j < 0)}.$$

Note that, by construction, $y_i \geq 0$ implies $\sqrt{\rho_i}U + \sqrt{1 - \rho_i}V_i \geq 0$ or $V_i \geq U\sqrt{\rho_i/(1 - \rho_i)}$, thus $\mathbb{P}(y_{j+1} \geq 0, y_i < 0 \text{ for all } i \leq j)$ can be simplified with

$$\begin{aligned} & \mathbb{P}(y_{j+1} \geq 0, y_i < 0 \text{ for all } i \leq j) \\ &= \mathbb{P}\left(V_{j+1} \geq -U\sqrt{\frac{\rho_{j+1}}{1 - \rho_{j+1}}}, V_i < -U\sqrt{\frac{\rho_i}{1 - \rho_i}} \text{ for all } i \leq j\right) \\ &= \int_{-\infty}^{\infty} \mathbb{P}\left(V_{j+1} \geq -U\sqrt{\frac{\rho_{j+1}}{1 - \rho_{j+1}}}, V_i < -U\sqrt{\frac{\rho_i}{1 - \rho_i}} \text{ for all } i \leq j \middle| u\right) \mathbb{P}_U(u) du \\ &= \int_{-\infty}^{\infty} \phi(u) \Phi\left(U\sqrt{\frac{\rho_{j+1}}{1 - \rho_{j+1}}}\right) \prod_{i=1}^j \Phi\left(-U\sqrt{\frac{\rho_i}{1 - \rho_i}}\right) du \\ &= \int_{-\infty}^{\infty} \phi(u) \left(1 - \Phi\left(-U\sqrt{\frac{\rho_{j+1}}{1 - \rho_{j+1}}}\right)\right) \prod_{i=1}^j \Phi\left(-U\sqrt{\frac{\rho_i}{1 - \rho_i}}\right) du. \end{aligned}$$

Similarly,

$$\mathbb{P}(y_i < 0 \text{ for all } i \leq j) = \int_{-\infty}^{\infty} \phi(u) \prod_{i=1}^j \Phi\left(-U\sqrt{\frac{\rho_i}{1 - \rho_i}}\right) du.$$

Combining these two, we get

$$\mathbb{P}(y_{j+1} \geq 0 | y_i < 0 \text{ for all } i \leq j) = 1 - \frac{\int_{-\infty}^{\infty} \phi(u) \Phi\left(-U\sqrt{\frac{\rho_{j+1}}{1 - \rho_{j+1}}}\right) \prod_{i=1}^j \Phi\left(-U\sqrt{\frac{\rho_i}{1 - \rho_i}}\right) du}{\int_{-\infty}^{\infty} \phi(u) \prod_{i=1}^j \Phi\left(-U\sqrt{\frac{\rho_i}{1 - \rho_i}}\right) du}.$$

Since, $\rho_i \leq 2(1 - \epsilon)/(2 - 2\epsilon + \epsilon^2)$ and $\sum_{i=1}^j \rho_i \leq 1/\epsilon^2$, the negative term in the above equation is maximized when $\rho_i = 2(1 - \epsilon)/(2 - 2\epsilon + \epsilon^2)$ for $(2 - 2\epsilon + \epsilon^2)\epsilon^2/2(1 - \epsilon)$ terms and 0 for the remaining terms. **Adel: But then $\sum \rho_i = \epsilon^2$** Note that this is not a function of n rather, if $S \sim \mathcal{N}_{k+1}(\mathbf{0}, k_1 \mathbf{I} + k_2 \mathbf{J})$, where $k = \lceil (2 - 2\epsilon + \epsilon^2)\epsilon^2/2(1 - \epsilon) \rceil$, $k_2 = 2(1 - \epsilon)/(2 - 2\epsilon + \epsilon^2)$ and $k_1 = 1 - k_2$. Then

$$\mathbb{P}(y_{j+1} \geq 0 | y_i < 0 \text{ for all } i \leq j) \leq \mathbb{P}(S_{k+1} \geq 0 | S_i < 0 \text{ for all } i \leq k),$$

where k, k_1, k_2 are constants that depend only on ϵ . Thus

$$C_z = 1 - \mathbb{P}(S_{k+1} \geq 0 | S_i < 0 \text{ for all } i \leq k),$$

satisfies the requirements.

B.6 Proof of Lemma 6

We have $\mathbf{A} = \mathbf{I} - \rho \mathbf{W}$, and $\|\mathbf{W}\|_\infty \leq 1$ and $|\rho| \leq 1 - \epsilon$, thus spectral radius of $\rho \mathbf{W}$ is less than $1 - \epsilon$. This implies $\mathbf{A} = \mathbf{I} - \rho \mathbf{W}$ is nonsingular and

$$\mathbf{A}^{-1} = (\mathbf{I} - \rho \mathbf{W})^{-1} = \sum_{k=0}^{\infty} \rho^k \mathbf{W}^k$$

Using sub-multiplicative property of the infinity norm, $\|\mathbf{W}\|_\infty \leq 1$ implies $\|\mathbf{W}^k\|_\infty \leq 1$ for any $k \geq 1$. With this we get

$$\|\mathbf{A}^{-1}\|_\infty = \left\| \sum_{k=0}^{\infty} \rho^k \mathbf{W}^k \right\|_\infty \leq \sum_{k=0}^{\infty} |\rho|^k \|\mathbf{W}^k\|_\infty \leq \sum_{k=0}^{\infty} |\rho|^k = \frac{1}{1 - |\rho|} \leq \frac{1}{\epsilon}.$$

Consequently, $\|\mathbf{A}^{-2}\|_\infty \leq (\|\mathbf{A}^{-1}\|_\infty)^2 \leq 1/\epsilon^2$.

Next, we want to prove $(\mathbf{A}^{-2})_{ii} \geq 1$. Note that $(\mathbf{A}^{-1})_{ii} \geq 1$ implies $(\mathbf{A}^{-2})_{ii} \geq 1$, since $(\mathbf{A}^{-2})_{ii} = \mathbf{a}_i^\top \mathbf{a}_i = \|\mathbf{a}_i\|_2^2$ where $\mathbf{a}_i$ is the i -th row of $\mathbf{A}^{-1}$ and if $(\mathbf{A}^{-1})_{ii} \geq 1$ then $\|\mathbf{a}_i\|_2 \geq 1$.

Note that since $\mathbf{W}$ is a real symmetric matrix, we can write $\mathbf{W} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$, where $\mathbf{\Lambda}$ is the diagonal matrix containing the eigenvalues $\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ of $\mathbf{W}$ and $\mathbf{Q}$ is a orthogonal matrix of eigenvectors. Also, since $\mathbf{W}$ has diagonal elements zero and row sum bounded by 1, using Gershgorin circle theorem all eigenvalues of $\mathbf{W}$ are bounded by 1, specifically, $|\lambda_j| \leq 1$ for all $1 \leq j \leq n$.

Using this, we first show that $(\mathbf{W}^{2m})_{ii} \geq (\mathbf{W}^{2m+1})_{ii} \geq 0$. First, note that since $\mathbf{W}$ has all positive elements, thus $\mathbf{W}^k$ has all elements positive as well of all k , thus $(\mathbf{W}^{2m+1})_{ii} \geq 0$. Next, since $\mathbf{W} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$ implies $\mathbf{W}^k = \mathbf{Q} \mathbf{\Lambda}^k \mathbf{Q}^\top$, thus for all k ,

$$(\mathbf{W}^k)_{ii} = \mathbf{e}_i^\top \mathbf{Q} \mathbf{\Lambda}^k \mathbf{Q}^\top \mathbf{e}_i = \sum_{j=1}^n q_{ij}^2 \lambda_j^k.$$

Finally, since $|\lambda_j| \leq 1$, thus $\lambda_j^{2m} \geq \lambda_j^{2m+1}$, and consequently

$$(\mathbf{W}^{2m})_{ii} = \sum_{j=1}^n q_{ij}^2 \lambda_j^{2m} \geq \sum_{j=1}^n q_{ij}^2 \lambda_j^{2m+1} = (\mathbf{W}^{2m+1})_{ii}.$$

Since $(\mathbf{W}^{2m})_{ii} \geq (\mathbf{W}^{2m+1})_{ii} \geq 0$, if $\rho \geq 0$, $\rho^{2m}(\mathbf{W}^{2m})_{ii} + \rho^{2m+1}(\mathbf{W}^{2m+1})_{ii} \geq 0$. In the second case, when $\rho < 0$, we have $\rho^{2m}(\mathbf{W}^{2m})_{ii} \geq -\rho^{2m+1}(\mathbf{W}^{2m+1})_{ii}$, since $\rho^{2m} \geq -\rho^{2m+1} \geq 0$, and $\rho^{2m}(\mathbf{W}^{2m})_{ii} + \rho^{2m+1}(\mathbf{W}^{2m+1})_{ii} \geq 0$ and thus, we get $\rho^{2m}(\mathbf{W}^{2m})_{ii} + \rho^{2m+1}(\mathbf{W}^{2m+1})_{ii} \geq 0$.

We now rewrite $\mathbf{A}^{-1}$ as follows,

$$\begin{aligned} \mathbf{A}^{-1} &= \sum_{k=0}^{\infty} \rho^k \mathbf{W}^k = \sum_{m=0}^{\infty} (\rho^{2m} \mathbf{W}^{2m} + \rho^{2m+1} \mathbf{W}^{2m+1}) \\ &= \mathbf{I} + \rho \mathbf{W} + \sum_{m=1}^{\infty} (\rho^{2m} \mathbf{W}^{2m} + \rho^{2m+1} \mathbf{W}^{2m+1}). \end{aligned}$$

Thus, we have $(\mathbf{A}^{-1})_{ii} = 1 + \sum_{m=1}^{\infty} \{\rho^{2m}(\mathbf{W}^{2m})_{ii} + \rho^{2m+1}(\mathbf{W}^{2m+1})_{ii}\}$, since $\mathbf{I}$ has all diagonal elements 1 and $\mathbf{W}$ has all diagonal elements 0. We finally conclude the proof, since $\rho^{2m}(\mathbf{W}^{2m})_{ii} + \rho^{2m+1}(\mathbf{W}^{2m+1})_{ii} \geq 0$, we get $(\mathbf{A}^{-1})_{ii} \geq 1$.